

Pharmacology of anti-hepatitis drugs (วิชาเภสัชวิทยา 2)

รศ.ดร. วิลาสินี หิรัญพานิช ชำโตะ
ภาควิชาเภสัชวิทยา
คณะเภสัชศาสตร์ มหาวิทยาลัยมหิดล



What is hepatitis?

The A, B, C, D and E of hepatitis

There are five main hepatitis viruses



What vaccines are available for which types of hepatitis?

- ✓ Hepatitis A
- ✓ Hepatitis B
- ✗ Hepatitis C
- ✓ Hepatitis D
- ✓ Hepatitis E

Inflammation of the liver

Estimated cases world-wide (per year)

Hepatitis **B and C**:
400 million cases

Gradual death:

1.4 million people die worldwide from hepatitis every year

Treatment:

90% of hepatitis C patients can be healed within three to six months

How does the virus spread?



Hepatitis **A and E**:
Lack of food hygiene, contaminated water and sub-standard sanitary facilities



Hepatitis **B, C and D**:
Blood, sperm and other bodily fluids

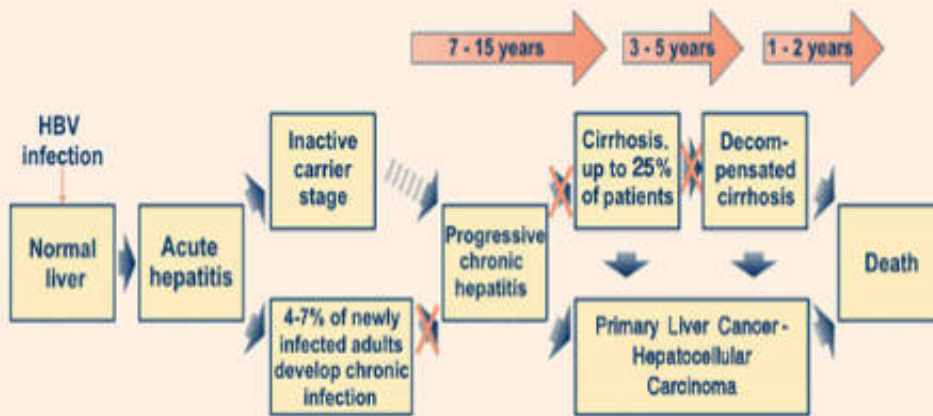
Source: WHO

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HBV and HCV are most common cause of chronic hepatitis, cirrhosis and hepatocellular carcinoma (HCC)

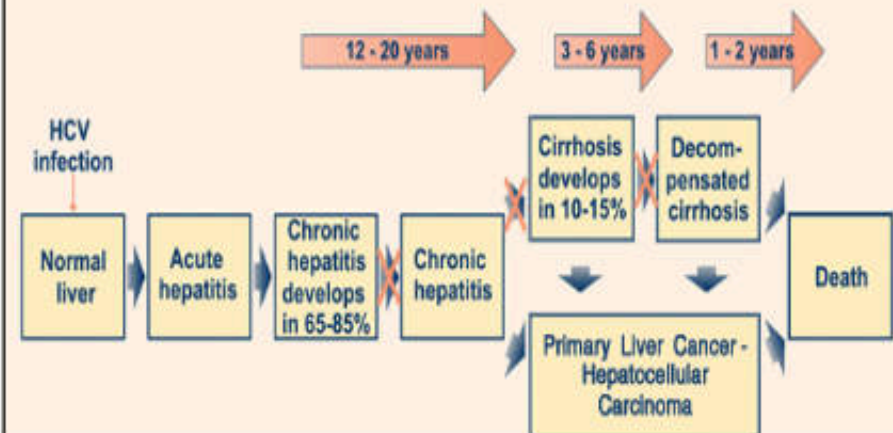
Natural History of Hepatitis B Infection

HBV



Natural History of Hepatitis C Infection

HCV

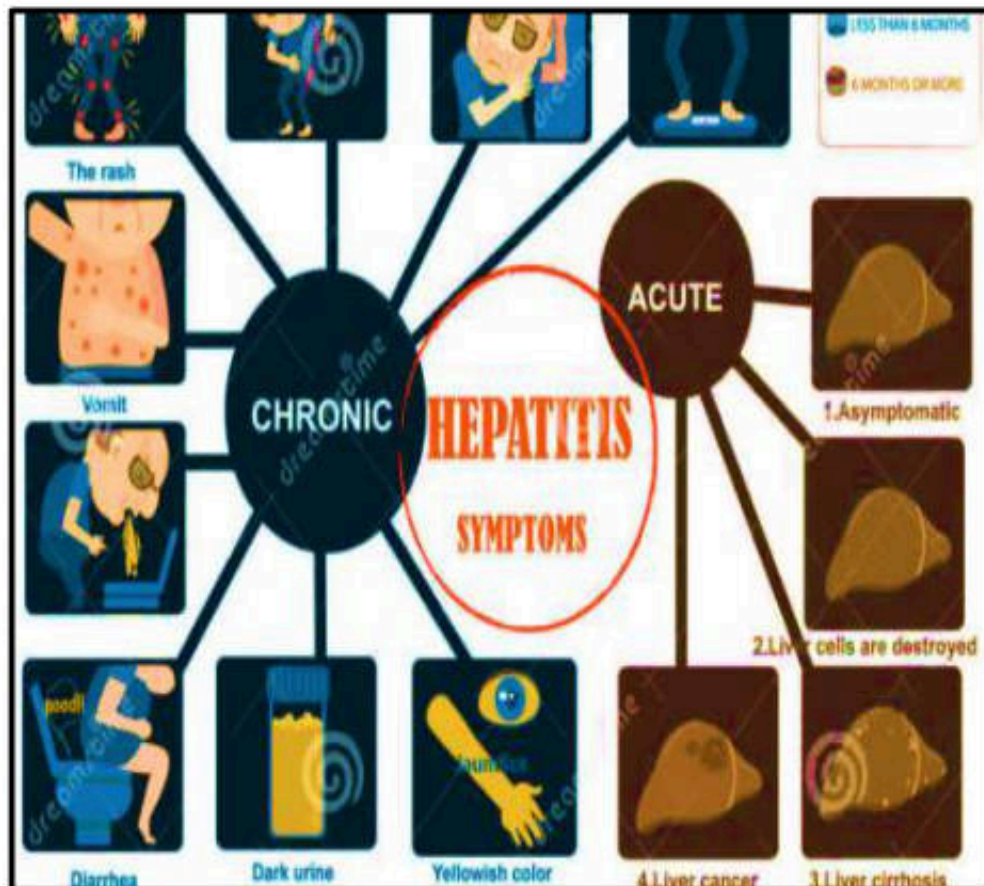


- DNA virus
- vaccine-preventable illness
- current treatments for HBV are not completely curative
- 15-40% risk of HCC (hepatocellular carcinoma)

- RNA virus
- no vaccine available to prevent HCV
- HCV can be cured with effective treatment
- 6-10% risk of HCC

Symptoms of hepatitis

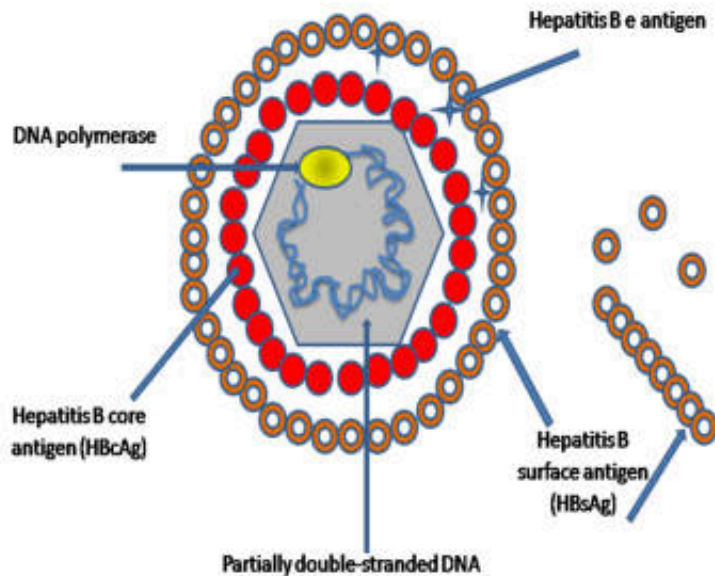
Pharmacological treatments of hepatitis



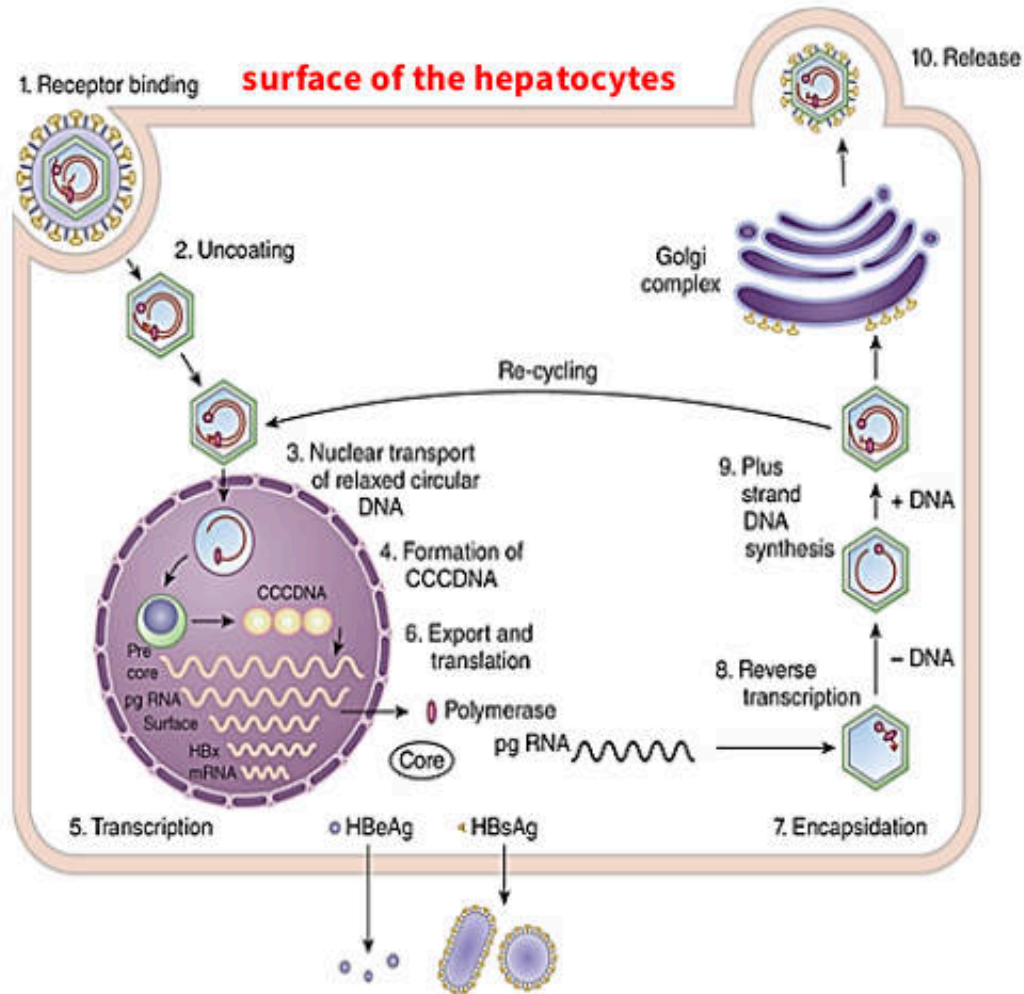
Type	Drugs
Hepatitis A virus	No
Hepatitis B virus	PegIFN/nucleoside and nucleotide analogues (NAs)
Hepatitis C virus	PegIFN/Direct Acting Antivirals (DAAs)
Hepatitis D virus	PegIFN- α
Hepatitis E virus	No

HBV Genome and Life Cycle

DNA Virus



cccDNA: covalently closed circular DNA



Goals of Hepatitis B Treatment

Acute HBV: no specific treatment , shortening the duration of the disease

- supportive care: maintaining comfort and adequate nutritional balance, including replacement of fluids lost from vomiting and diarrhea
- lowering the risk of chronicity

Chronic HBV: treated with medicines

Goal of treatments:

- Improve survival and quality of life by preventing disease progression, and consequently HCC development
- Prevent mother to child transmission, hepatitis B reactivation
- Prevention and treatment of HBV-associated extrahepatic manifestations

HBV Drug Targets

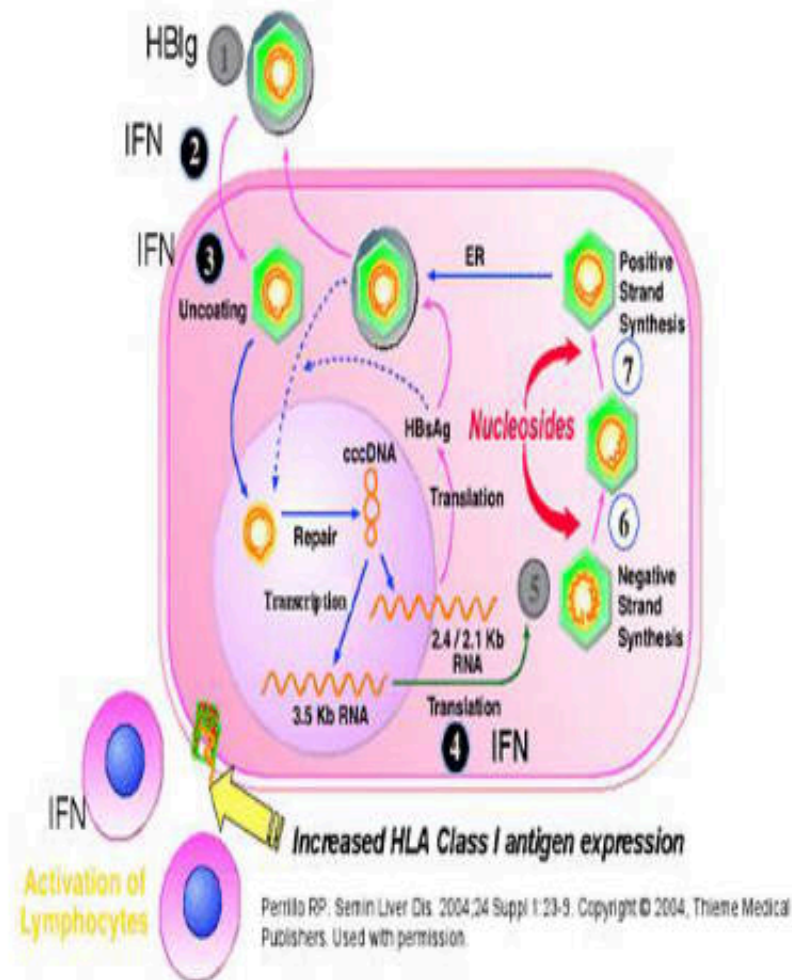
- Current HBV therapies cannot achieve a complete cure→ No drug can destroy cccDNA !!!!
- “Goals of chronic HBV therapy”
 1. Suppression of HBV DNA to undetectable levels
 2. Seroconversion of HBeAg (or more rarely, HBsAg) from positive to negative
 3. Reduction in elevated serum aminotransferase levels

1. Interferon (IFN)	2. Nucleoside/-tide analogues	
▪ Non-specific antiviral immunomodulatory ▪ Treatment is finite (48–52 weeks) ▪ High ADR/effect is not predictable.	▪ Inhibit the HBV polymerase ▪ Time of treatment is depend on individuals ▪ May require life long treatment ▪ Oral treatment / Low ADR ▪ High cost	▪ Adefovir dipivoxi (ADV) ▪ Lamivudine (LAM) ▪ Telbivudine (TBV) ▪ <u>Entecavir</u> (ETV) ▪ <u>Tenofovir disoproxil fumarate</u> (TDF) ▪ <u>Tenofovir alafenamide</u> (TAF)

1. Interferons

- Antiviral, immunomodulatory, and antiproliferative effects
- Cytokines which interfere with viral replication in host cells by inhibiting viral DNA synthesis
- Enhancing the cellular immune response against HBV-infected hepatocytes
- Available use for HBV treatment
- 1. Interferon (IFN) alfa-2b: inject 3 times/week
- 2. Pegylated interferon alfa-2_a (PegIFN)
 - Pegylation = polyethylene glycol attachment
 - Pegylation improves half-life and drug efficacy
 - Inject SC once-weekly
 - PegIFN- α 2a is a first-line agent approved by FDA for the treatment of HBV
 - Interferon- α 2b is approved for children 1 year or older

MECHANISMS OF ACTION



UNTOWARD EFFECTS

- **Acute influenza-like syndrome** (beginning several hours after injection): fever, chills, headache, myalgia, arthralgia, nausea, vomiting, and diarrhea
- **Depression**
- **Myelosuppression** with granulocytopenia and thrombocytopenia
- **Neurotoxicity**: somnolence, confusion, behavioral disturbance, and rarely, seizures
- **Autoimmune disorders**: thyroiditis and hypothyroidism
- **Uncommon ADRs**: cardiovascular effects with hypotension and tachycardia
- **Others: Elevations in hepatic enzymes** and triglycerides, alopecia, proteinuria and azotemia, interstitial nephritis, autoantibody formation, pneumonia, and hepatotoxicity

Pegylated IFNs are generally better tolerated than standard IFNs

Contraindications and cautions for IFN therapy for Chronic hepatitis B

- **Thyroid function** and hepatic enzymes should be monitored during IFN therapy
- **Should not** be administered in pregnancy: abortifacient in primates

Table 5 • Contraindications for Interferon- α Therapy for Chronic Hepatitis B

Hepatic decompensation :	- Bilirubin $>51.3 \mu\text{M}$ - Albumin $<30 \text{ g/L}$ - Prolonged prothrombin time $> 3 \text{ s}$
Portal hypertension :	- Variceal bleed - Ascites - Encephalopathy
Hypersplenism :	- Leukopenia ($<2 \times 10^9/\text{L}$) - Thrombocytopenia ($<7 \times 10^7/\text{L}$)
Renal impairment :	- Creatine $> 176.8 \mu\text{M}$
Autoimmune disease :	- Polyarteris nodosa - Rheumatoid arthritis
Severe major system dysfunction :	- Cardiac failure - Obstructive airways disease - Uncontrolled diabetes
Psychiatric :	- Severe depression - Suicide attempt
Addictive habits:	- Alcoholism - Current intravenous drug abuse

Ex. of drug-drug interactions

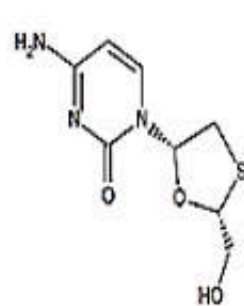
increased theophylline and methadone level

+didanosine: hepatic failure

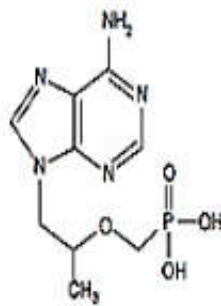
+zidovudine: cytopenia

2. Nucleoside/nucleotide analogs (NA)

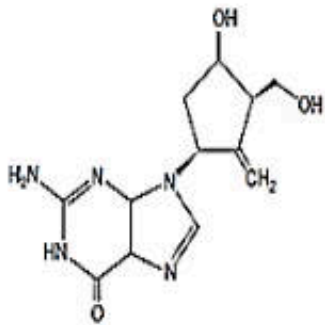
Directly inhibit HBV DNA polymerase →
~~reverse transcriptase~~ → ~~HBV DNA~~
 synthesis



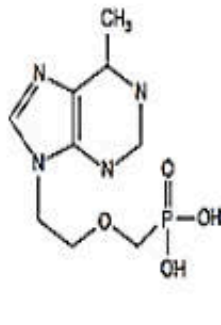
LAMIVUDINE



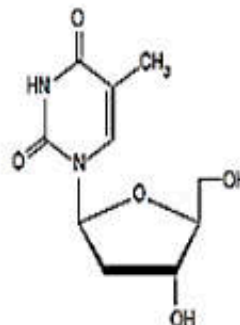
TENOFOVIR



ENTECAVIR



ADEFOVIR



TELBIVUDINE

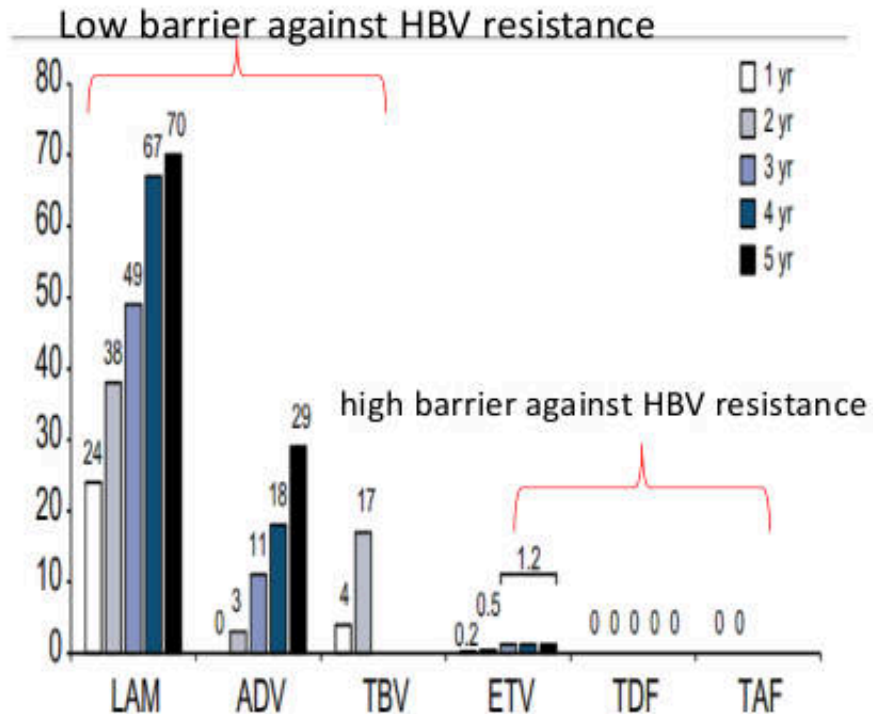


Fig. 3. Cumulative incidence of HBV resistance for lamivudine (LAM), adefovir (ADV), entecavir (ETV), telbivudine (TBV), tenofovir (TDF) and tenofovir alafenamide (TAF) in pivotal trials in nucleos(t)ide-naïve patients with chronic hepatitis B. (Collation of currently available data - not from head-to-head studies). No evidence of resistance has been shown after 8 years of TDF treatment.⁶⁹

ENTECAVIR

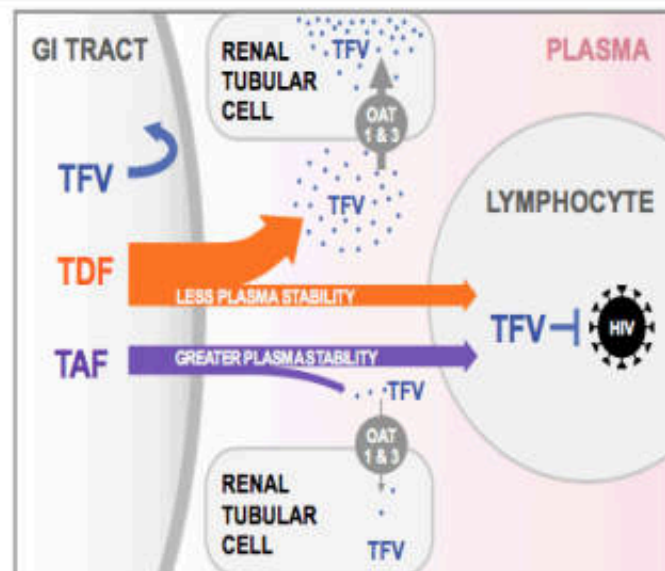
- Competitively inhibit 3 functions of HBV DNA polymerase:
 - base priming
 - reverse transcription of the negative strand
 - synthesis of the positive strand of HBV DNA
- Oral bioavailability 100% but is decreased by food
- Plasma $T_{1/2}$: 128–149 hrs. once-daily dosing
- Excreted by the kidney → dosage should be adjusted in renal insufficiency
 - Co-administration of entecavir with drugs that reduce renal function
- Higher barrier to the emergence of resistance
 - resistance is more frequent in lamivudine-refractory patients (50% in 5 years)
- Well tolerated: potential adverse events: headache, fatigue, dizziness, nausea, and upper abdominal pain

TENOFOVIR

- Tenofovir disoproxil fumarate (TDF) and tenofovir alafenamide (TAF)
- TAF: orally bioavailable prodrug of tenofovir (less toxicity)
 - Enhance delivery of active drug (TFV-DP) to hepatocytes
 - Less effect on renal dysfunction and bone mass density (BMD)
- **High efficacy and barrier of resistance:** but efficacy is lower in patients who have resistance to adefovir and double mutations (A181T/V and N236T)
- **Most common ADRs:**
 - nausea, abdominal pain, diarrhea, dizziness, and fatigue, Chronic renal insufficiency, decreases in bone mineral density and **Fanconi's syndrome** (*disorder of the kidney tubes*)

Prodrug Pharmacology

Tenofovir Disoproxil Fumarate and Tenofovir Alafenamide



TAF 25 mg results in 80–91% lower TFV plasma levels

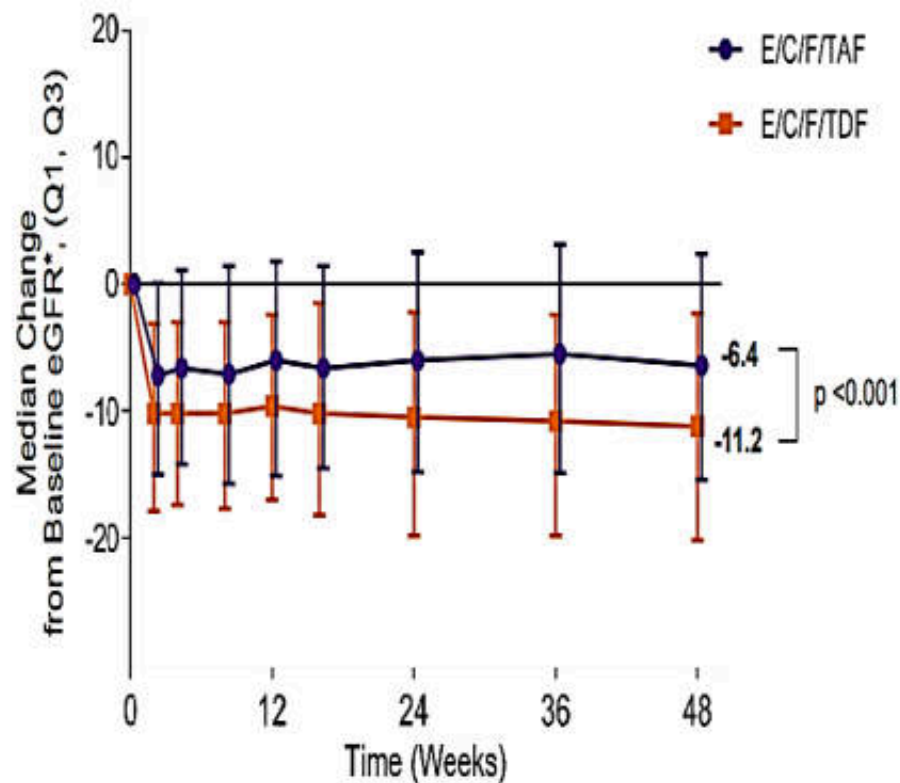
OAT, organic anion transporter; TAF, tenofovir alafenamide; TDF, tenofovir disoproxil fumarate; TFV, tenofovir; 1. Sax P, et al. JAIDS 2014; 67:52–58; 2. Sax P, et al. Lancet 2015;385:2608–15; 3. Data on file.

Background

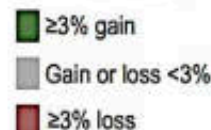
- Tenofovir disoproxil fumarate (TDF) is a widely used antiretroviral HIV, which has been associated with increased risk of CKD in col studies including D:A:D¹⁻³
- In Study 109, tenofovir alafenamide (TAF) has an improved renal safety profile compared to TDF through 2 years⁴
- Efficacy and renal outcomes by CKD risk category in virologically suppressed adults treated with E/C/F/TAF or a TDF-based regimen 96 weeks are presented

Tenofovir alafenamide Vs tenofovir disoproxil fumarate, co-formulated with elvitegravir, cobicistat, and emtricitabine, for initial treatment of HIV-1 infection: two randomized, double-blind, phase 3, non-inferiority trials

Kidneys



BMD Change

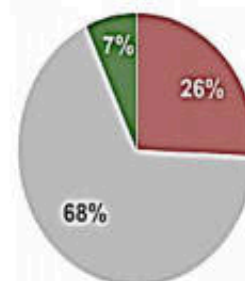


E/C/F/TAF (N=845)

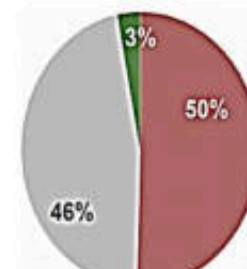
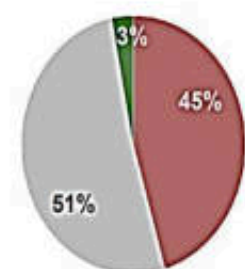
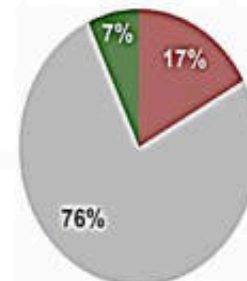
E/C/F/TDF (N=850)

Bone

Spine



Hip



* $p < 0.001$ for treatment-arm comparisons for each category

ยาที่ใช้ในการรักษาไวรัส HBV (Thai guideline)

1) Tenofovir alafenamide (TAF)



ขนาด 25 mg 1 เม็ด
วันละครั้ง



ควรเปลี่ยนยาหากมีภาวะไตวาย
เรื้อรัง (eGFR
<15mL/min/1.73 m²)



ใช้ในผู้ป่วยรายเดิมที่เคยได้รับการ
รักษาด้วย Lamivudine หรือ
Entecavir หรือ TDF

2) Tenofovir disoproxil fumarate (TDF)

- ใช้เป็นยาขนานแรกในหญิงตั้งครรภ์
- พิจารณาใช้ในผู้ที่มีค่า eGFR < 15 mL/min/1.73 m² และยังไม่ได้รับการบำบัดทดแทนไต และไม่สามารถใช้ยา Entecavir ได้

eGFR(mL/min/1.73 m ²)	ขนาดยา TDF
≥ 50	300 มิลลิกรัมทุก 24 ชั่วโมง
30-49	300 มิลลิกรัมทุก 48 ชั่วโมง
10-29	300 มิลลิกรัมทุก 72-96 ชั่วโมง
Hemodialysis	300 มิลลิกรัมทุก 7 วัน

Monitoring of patients treated with ETV, TDF, TAF

“Should monitor ALT and serum HBV DNA in all patients”

- Patients at risk of renal disease treated with any NAs and all patients treated with TDF:
 - renal monitoring : eGFR
 - serum phosphate levels (q. 3 mon.)
- Patients on TDF at risk of development **renal or bone** disease should be considered for switch to ETV or TAF

Table 5. Indications for selecting ETV or TAF over TDF.*

1. Age >60 years

2. Bone disease

Chronic steroid use or use of other medications that worsen bone density

History of fragility fracture

Osteoporosis

3. Renal alteration**

eGFR <60 ml/min/1.73 m²

Albuminuria >30 mg/24 h or moderate dipstick proteinuria

Low phosphate (<2.5 mg/dl)

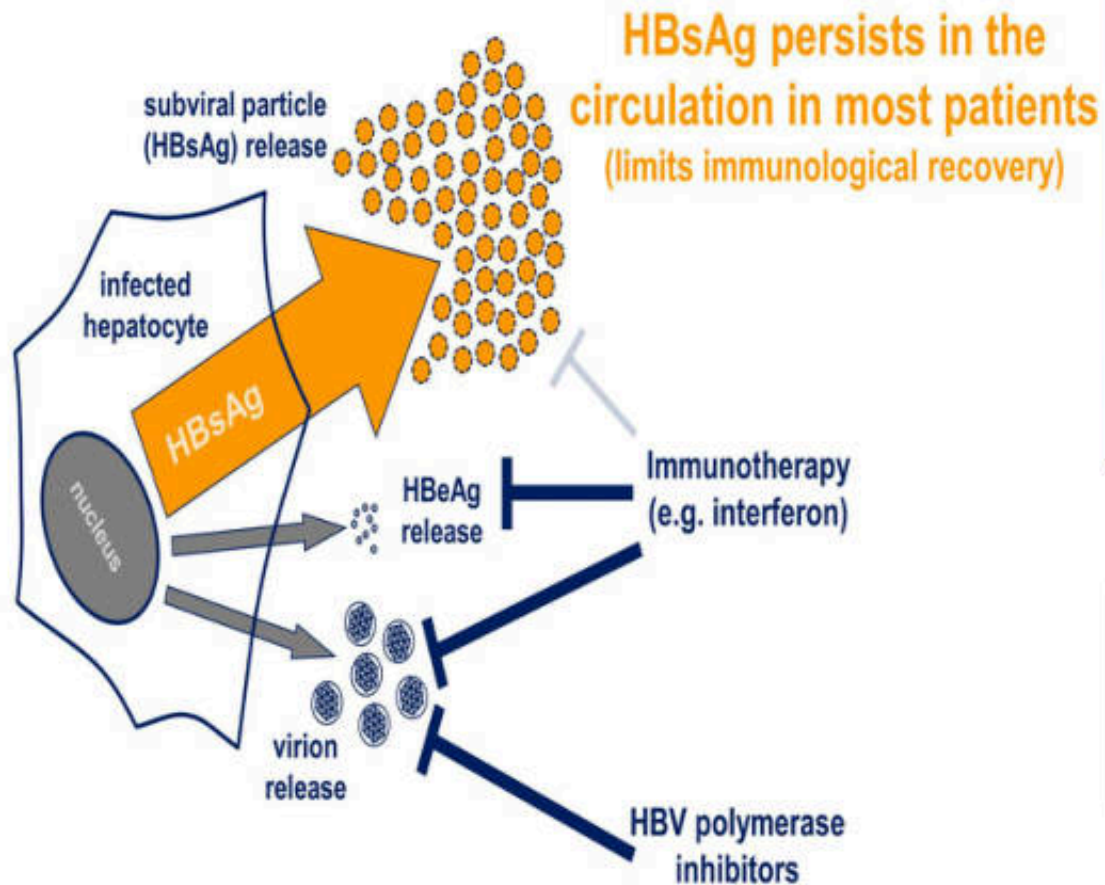
Hemodialysis

* TAF should be preferred to ETV in patients with previous exposure to nucleoside analogues.

** ETV dose needs to be adjusted if eGFR <50 ml/min; no dose adjustment of TAF is required in adults or adolescents (aged at least 12 years and of at least 35 kg body weight) with estimated creatinine clearance (CrCl) ≥15 ml/min or in patients with CrCl <15 ml/min who are receiving haemodialysis.

	ADEFOVIR DIPIVOXIL	LAMIVUDINE	TELIVUDINE
Mechanism of action	competitively inhibits HBV DNA polymerase and causes chain termination after incorporation into viral DNA	competing with deoxycytidine triphosphate for incorporation into the viral DNA, resulting in chain termination	phosphorylated compound competitively inhibits HBV DNA polymerase, resulting in incorporation into viral DNA and chain termination
Pharmacokinetics	Oral bioavailability: 59%, Protein binding is low (<5%), T _{1/2} : 5-18 h (once-daily dosing), excreted by renal (requires dose adjustment for renal dysfunction)	safely administered to patients with decompensated liver disease	Oral bioavailability: unaffected by food, low plasma protein binding (3%), T _{1/2} : 15 h and excretion by renal.
Resistance	Emergence of resistance is up to 29% after 5 years of use.	- lamivudine-resistant HBV isolates (15–30% of patients at 1 year, 65% after 5 years of therapy) - Cross-resistance with emtricitabine or entecavir	High resistance, (22% in 1 yr.), not effective in lamivudine-resistant HBV
AEs	well tolerated, but increase in serum creatinine	excellent safety profile	Mild (peripheral neuropathy, myopathy)
Teratogenic effect	embryo toxic in rats at high doses and is genotoxic in preclinical studies.	preventing vertical transmission of HBV from mother to newborn when given in the last 4 weeks of gestation	2 nd and 3 rd trimester of pregnancy in HBeAg+ mothers can safely reduce perinatal HBV transmission rates

Limitation of current HBV antiviral therapies



De novo combination therapy with two NAs with high barrier to resistance (ETV, TDF, TAF) is not recommended

“Theoretically”

combined NA +

PegIFN α

→ combining the potent antiviral effect of NA

+ the immune modulation of IFN α

“BUT”

The evidences of superiority of a combined approach, is lacking



De novo combination of NA + PegIFN α is not recommended

Comparison of current treatment chronic hepatitis B according to EASL 2017 guideline

Features	PegIFN α	ETV, TDF, TAF
Route of administration	Subcutaneous injections	Oral
Treatment duration	48 weeks	Long-term until HBsAg loss (stopping NA after some years might be considered in selected cases) ¹
Tolerability	Low	High
Long-term safety concerns	Very rarely persistence of on-treatment adverse events (psychiatric, neurological, endocrinological)	Probably not (uncertainties regarding kidney function, bone diseases for some NA)
Contraindications	Many (i.e., decompensated disease, co-morbidities etc.)	None (dose adjustment according to eGFR ²)
Strategy	Induction of a long-term immune control by finite treatment	Stopping hepatitis and disease progression by inhibiting viral replication
Level of viral suppression	Moderate (variable response pattern)	Universally high
Effect on HBeAg loss	Moderate, depending on baseline characteristics	Low in the first year, increases to moderate during long-term treatment
Effect on HBsAg levels	Variable, depending on baseline characteristics (overall higher as compared to NA)	Low: slowly increases with treatment time in HBeAg-positive patients ³ ; usually very low in HBeAg-negative patients
Risk of relapse after treatment cessation	Low for those with sustained response 6–12 months after therapy	Moderate if consolidation treatment provided after HBeAg seroconversion. High for HBeAg-negative disease
Early stopping rules	Yes	No
Risk of viral resistance development	No	Minimal to none ⁴

PegIFN α , pegylated interferon alfa; ETV, entecavir; TDF, tenofovir disoproxil fumarate; TAF, tenofovir alafenamide; NA, nucleoside/nucleotide analogues; eGFR, estimated glomerular filtration rate.

¹ See section on 'Treatment strategies'.

² Dose adjustments in patients with eGFR <50 ml/min are required for all NA, except for TAF (no dose recommendation for TAF in patients with CrCl <15 ml/min who are not receiving haemodialysis).

³ A plateau in serologic responses has been observed beyond treatment year 4.

⁴ So far no TDF or TAF resistance development has been detected.

Approved Antiviral Therapies in Adults and Children

Drug	Dose in Adults*	Use in Children*	Pregnancy Category ^f	Potential Side Effects [†]	Monitoring on Treatment [‡]
Preferred					
Peg-IFN- α -2a (adult) IFN- α -2b (children)	180 mcg weekly	≥ 1 year dose: 6 million IU/m ² three times weekly [§]	C	Flu-like symptoms, fatigue, mood disturbances, cytopenia, autoimmune disorders in adults, anorexia and weight loss in children	Complete blood count (monthly to every 3 months) TSH (every 3 months) Clinical monitoring for autoimmune, ischemic, neuropsychiatric, and infectious complications
Entecavir	0.5 mg daily	≥ 2 years dose: weight-based to 10-30 kg: 0.5 mg daily ≥ 12 years	C	Lactic acidosis (decompensated cirrhosis only)	Lactic acid levels if there is clinical concern Test for HIV before treatment initiation
Tenofovir dipovoxil fumarate	300 mg daily	≥ 12 years	B	Nephropathy, Fanconi syndrome, osteomalacia, lactic acidosis	Creatinine clearance at baseline If at risk for renal impairment, creatinine clearance, serum phosphate, urine glucose, and protein at least annually Consider bone density study at baseline and during treatment in patients with history of fracture or risks for osteopenia Lactic acid levels if there is clinical concern Test for HIV before treatment initiation
Tenofovir alafenamide	25 mg daily	—	There are insufficient human data on use during pregnancy to inform a drug-associated risk of birth defects and miscarriage.	Lactic acidosis	Lactic acid levels if clinical concern Assess serum creatinine, serum phosphorus, creatinine clearance, urine glucose, and urine protein before initiating and during therapy in all patients as clinically appropriate Test for HIV before treatment initiation

Approved Antiviral Therapies in Adults and Children (cont.)

(Non-preferred)

Lamivudine	100 mg daily	≥2 years dose: 3 mg/kg daily to max 100 mg	C	Pancreatitis Lactic acidosis	Amylase if symptoms are present Lactic acid levels if there is clinical concern
Adefovir	10 mg daily	≥12 years	C	Acute renal failure Fanconi syndrome Lactic acidosis	Test for HIV before treatment initiation Creatinine clearance at baseline If at risk for renal impairment, creatinine clearance, serum phosphate, urine glucose, and urine protein at least annually Consider bone density study at baseline and during treatment in patients with history of fracture or risks for osteopenia
Telbivudine	600 mg daily	—	B	Creatine kinase elevation and myopathy Peripheral neuropathy Lactic acidosis	Lactic acid levels if clinical concern Creatine kinase if symptoms are present Clinical evaluation if symptoms are present Lactic acid levels if there is clinical concern

*Dose adjustments are needed in patients with renal dysfunction.

[†]In 2015, the U.S. Food and Drug Administration replaced the pregnancy risk designation by letters A, B, C, D, and X with more specific language on pregnancy and lactation. This new labeling is being phased in gradually, and to date only TAF includes these additional data.

[‡]Per package insert.

[§]Peg-IFN- α -2a is not approved for children with chronic hepatitis B, but is approved for treatment of chronic hepatitis C. Providers may consider using this drug for children with chronic HBV. The duration of treatment indicated in adults is 48 weeks.

^{||}Entecavir dose is 1 mg daily if the patient is lamivudine experienced or if they have decompensated cirrhosis.

Abbreviation: TSH, thyroid stimulating hormone.

New biomarkers of HBV infection and Future treatment options

New biomarkers

- *Viral cccDNA, Hepatitis B core-related antigen (HBcrAg), Circulating HBV RNA*

Now there is no true HBV 'cure'

Future treatment options

1. Direct antivirals

- HBV entry inhibitors → aiming at cccDNA destruction or silencing

2. Immunotherapeutic agents

- Toll-like receptors 7 (TLR7) agonists → to restore innate immunity in CHB patient

ข้อเสนอแนะ

1. พิจารณาใช้ยา pegylated interferon ในผู้ป่วยที่มีระดับไวรัส ALT มากกว่าหรือเท่ากับ 2 เท่าของค่าปกติ ระดับไวรัสต่ำ (น้อยกว่า $8 \log \text{ IU/mL}$ ในผู้ป่วย HBeAg บวกหรือน้อยกว่า $6 \log \text{ IU/mL}$ ในผู้ป่วย HBeAg ลบ แต่ไม่ควรใช้ ในผู้ป่วยตับแข็งที่เป็นมากแล้ว (decompensated cirrhosis)
2. กรณีเลือกใช้ยาชนิดรับประทาน ให้พิจารณาใช้ entecavir หรือ tenofovir เป็นตัวแรกเพื่อลดความเสี่ยงต่อการื้อยาในระยะยาว

โครงการรณรงค์ป้องกันและแก้ไข
โรคเอดส์และโรคติดต่อทางเพศสัมพันธ์
สำหรับแพทย์เวชปฏิบัติทั่วไป
พ.ศ. 2567

สมาคมธุรกิจการบินไทย-จีน
ปี ๒๕-๖ ซึ่งอยู่ในประเทศไทย ปี 2558

Thailand Practice Guideline
for Management of ChronicHepatitis
B and C 2015

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mailto:frank@cs.cmu.edu

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กระทรวงสาธารณสุข
Ministry of Public Health

Antihepatitic used in THailand

Drugs	Dose
Conventional interferon α	ขนาด 5-10 MU ฉีดเข้าชั้นใต้ผิวหนัง 3 ครั้ง ต่อสัปดาห์ เป็นเวลานาน 6 เดือน
Pegylated interferon α -2a	ฉีดเข้าชั้นใต้ผิวหนังขนาด 180 ไมโครกรัม สัปดาห์ละ 1 ครั้ง เป็นเวลานาน 12 เดือน
Pegylated interferon α -2b	ฉีดเข้าชั้นใต้ผิวหนัง ขนาด 1.5 ไมโครกรัมต่อกก. สัปดาห์ละ 1 ครั้ง เป็นเวลานาน 12 เดือน
Lamivudine	ขนาด 100-150 มิลลิกรัม รับประทานวันละ 1 ครั้ง
Adefovir dipivoxil	ขนาด 10 มิลลิกรัม รับประทานวันละ 1 ครั้ง
Entecavir	ขนาด 0.5 มิลลิกรัม รับประทานวันละ 1 ครั้ง
Telbivudine	ขนาด 600 มิลลิกรัม รับประทานวันละ 1 ครั้ง
Tenofovir disoproxil fumarate	ขนาด 300 มิลลิกรัม รับประทานวันละ 1 ครั้ง
Clevudine	ขนาด 30 มิลลิกรัม รับประทานวันละ 1 ครั้ง

Quiz

1. Why interferon is no longer first-line ?

- A. Interferon is ineffective against hepatitis viruses
- B. Interferon causes viral resistance
- C. Oral antivirals are more effective and better tolerated
- D. Interferon is no longer available worldwide

2. Why Tenofovir alafenamide (TAF) is preferred over tenofovir disoproxil (TDF)?

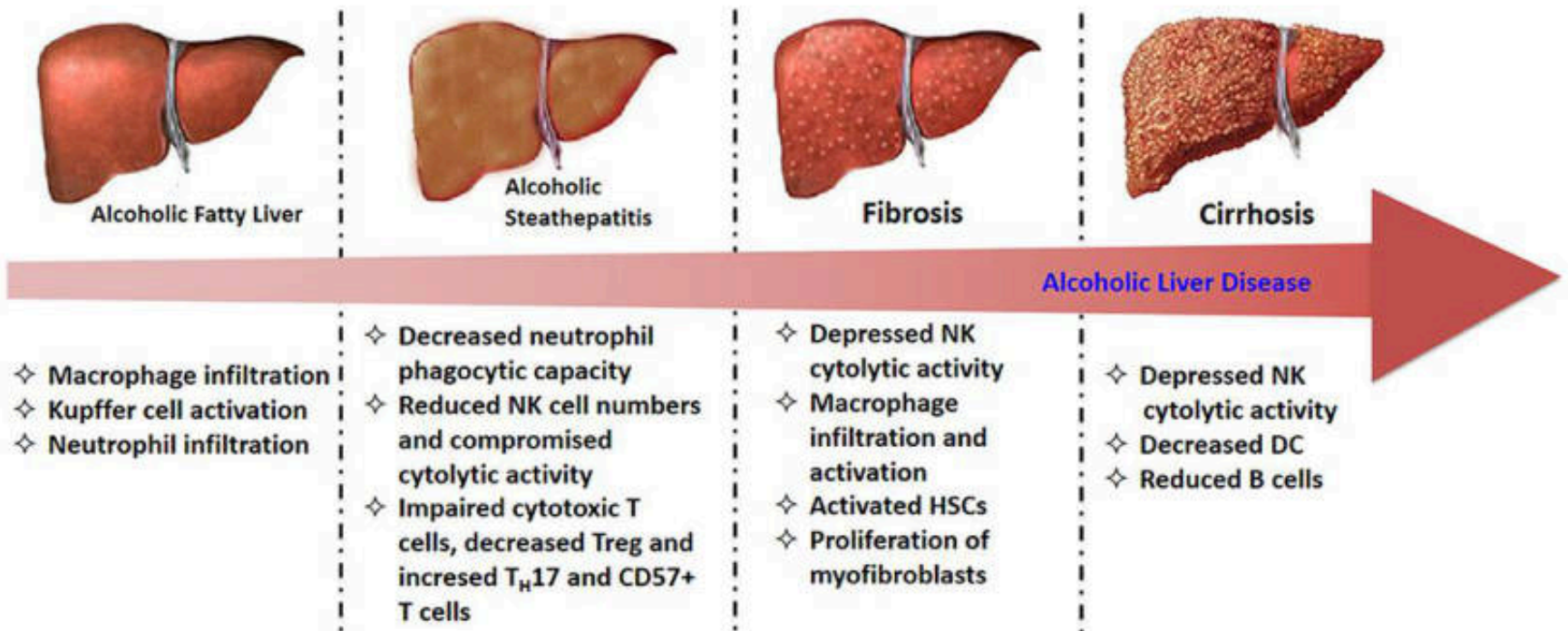
- A. stronger antiviral activity
- B. shorter half-life
- C. less renal and bone toxicity
- D. does not require renal dose adjustment

Treatment of hepatitis C virus (HCV)

Test-and-Treat

DAA-based therapy

Goal: End of HCV in 2573



Hepatitis C Virus (HCV)

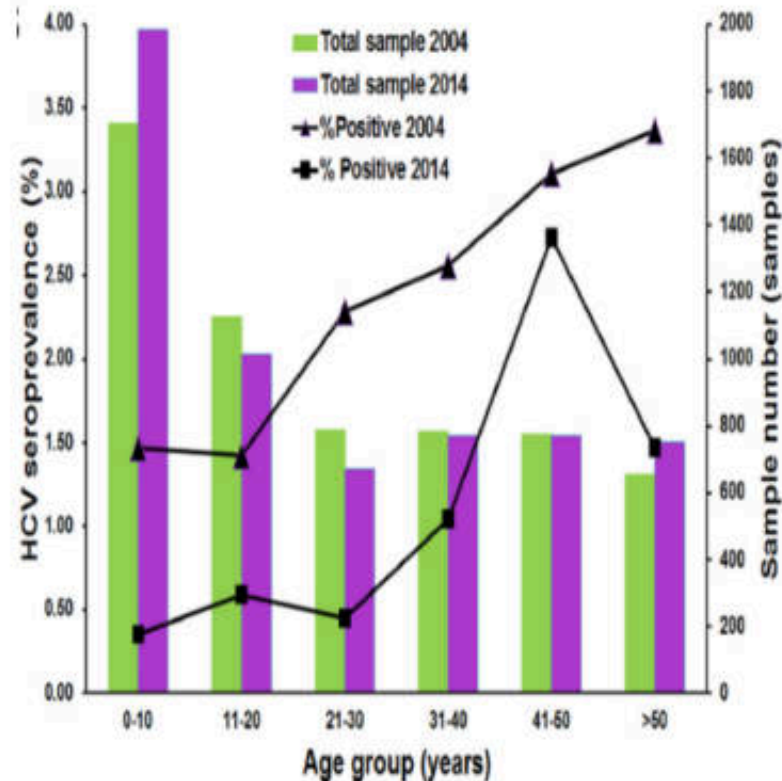
~ 71 million chronically infected with HCV worldwide

85% of infected patients will develop chronic infection, fibrosis and cirrhosis with or without hepatocellular carcinoma (HCC)

Exhibits remarkable within- and between-subject genetic heterogeneity

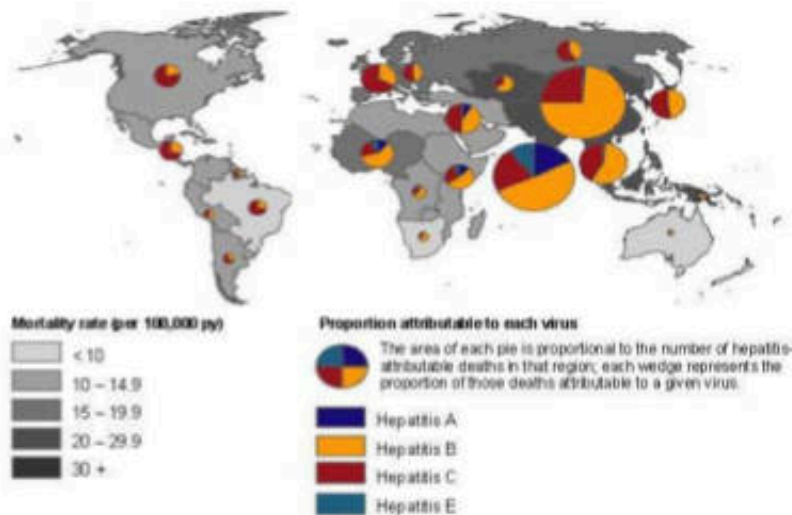
- No preventive vaccine

6 genotypes: genotype 1>3>2, 4, 5, 6

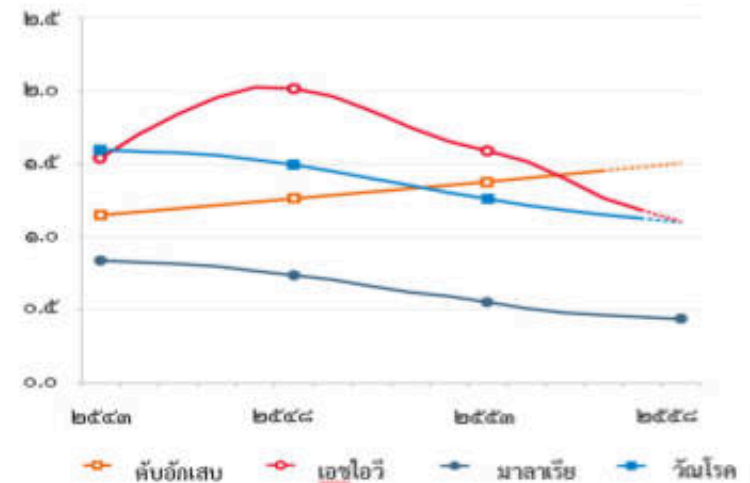


ความชุกของผู้ติดเชื้อ HCV ในไทย ปี พ.ศ. 2557

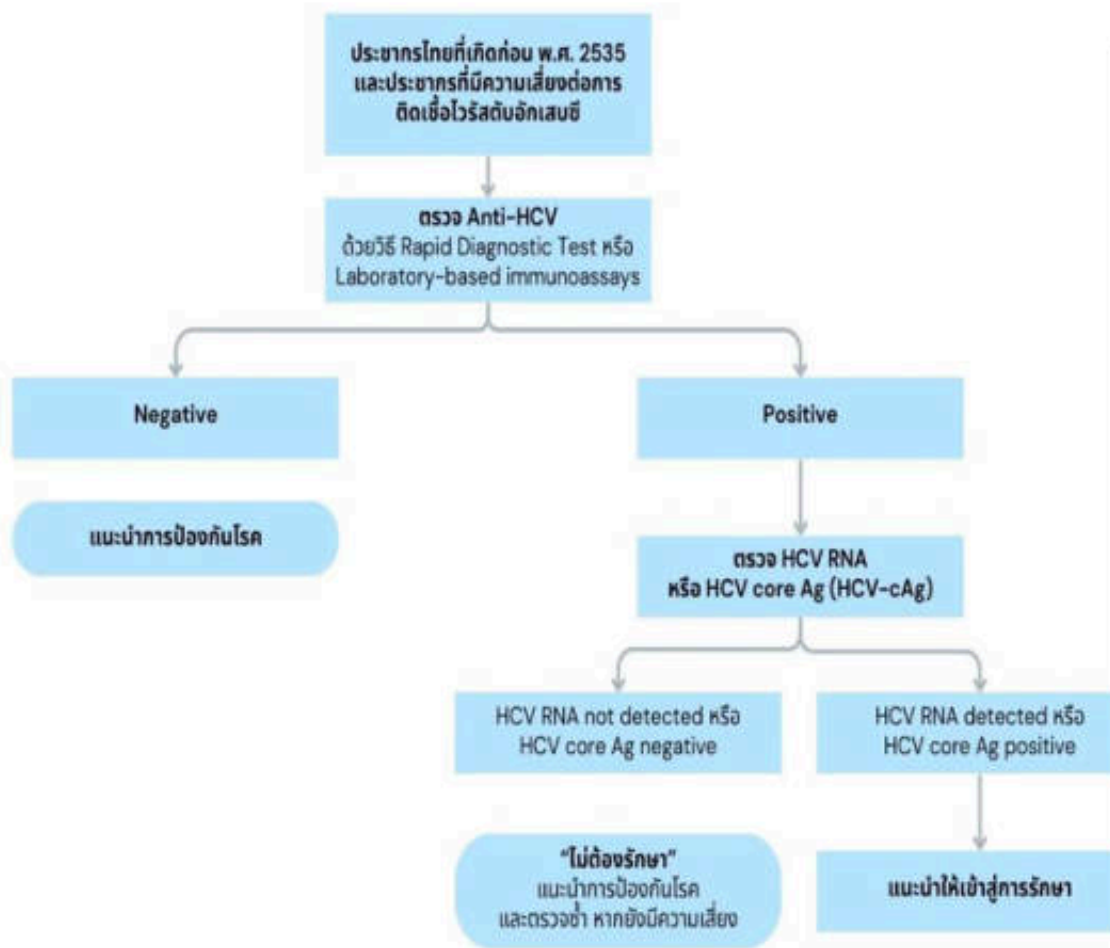
ประมาณจำนวนผู้เสียชีวิตทั่วโลกจากเชื้อไวรัสตับอักเสบ เชื้อเอชไอวี โรคมาลาเรีย และวัณโรค ปี พ.ศ. 2543 - 2558



จำนวนผู้เสียชีวิต (ล้านคน)



การตรวจคัดกรอง HCV

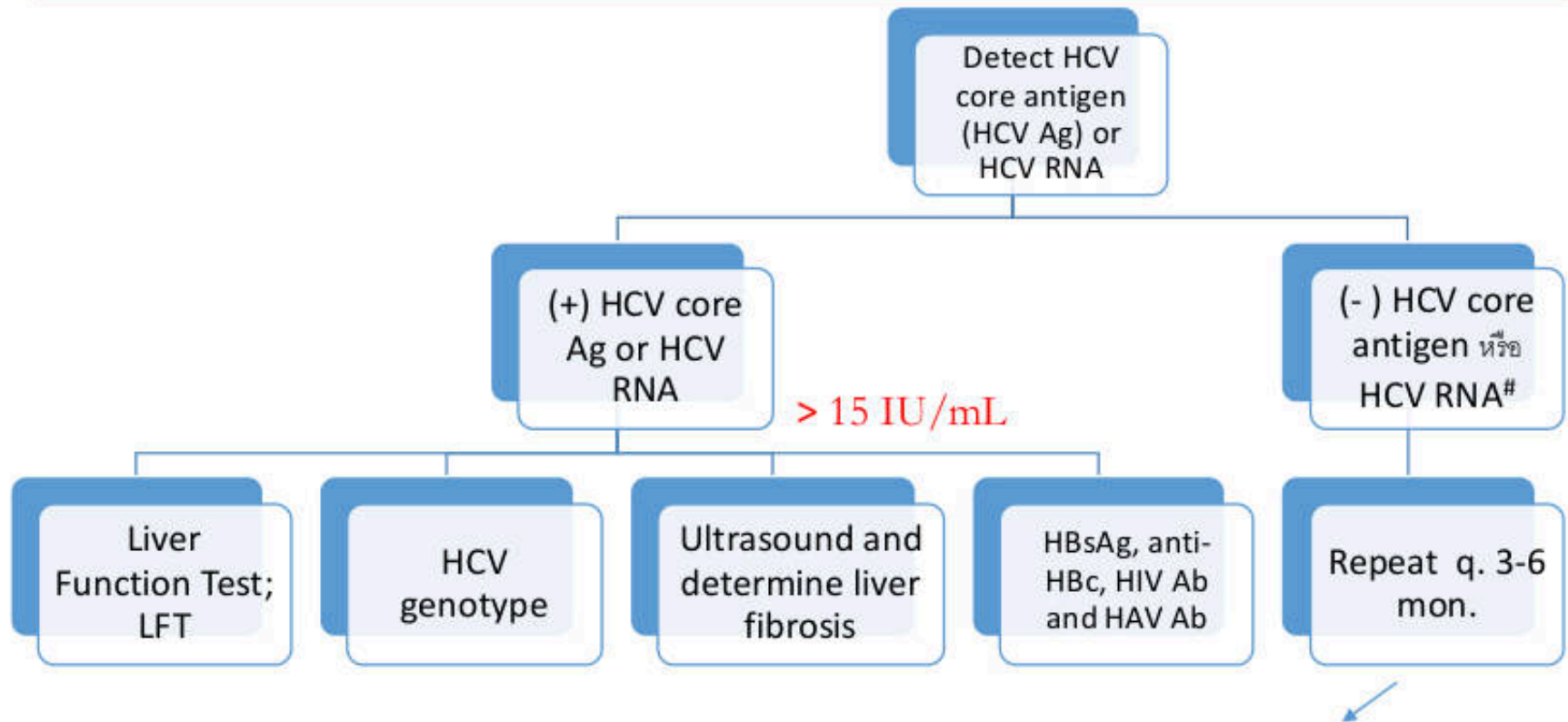


กลุ่มคนที่ได้สิทธิตรวจคัดกรองฟรี

- ประชากรไทยที่เกิดก่อนปี พ.ศ. 2535
- ประชากรไทยกลุ่มเสี่ยง 5 กลุ่ม
 - ติดเชื้อ HIV
 - ผู้ใช้ยาเสพติดชนิดฉีด
 - กลุ่มชายมีเพศสัมพันธ์กับชาย
- บุคลากรทางการแพทย์
- ผู้ต้องขัง

หมายเหตุ ในกรณีตรวจพบ HCV core Ag negative แต่มีผลการทำงานของค่าตับผิดปกติโดยไม่ทราบสาเหตุ แนะนำให้ตรวจยืนยันด้วย HCV RNA

Diagnosis ผู้ป่วยที่ตรวจซีรัม anti-HCV ให้ผลบวก



Note: Should determine for HCV RNA in patients with acute hepatitis C and high risk groups

Goals and endpoints of HCV therapy

- **Primary Goals**

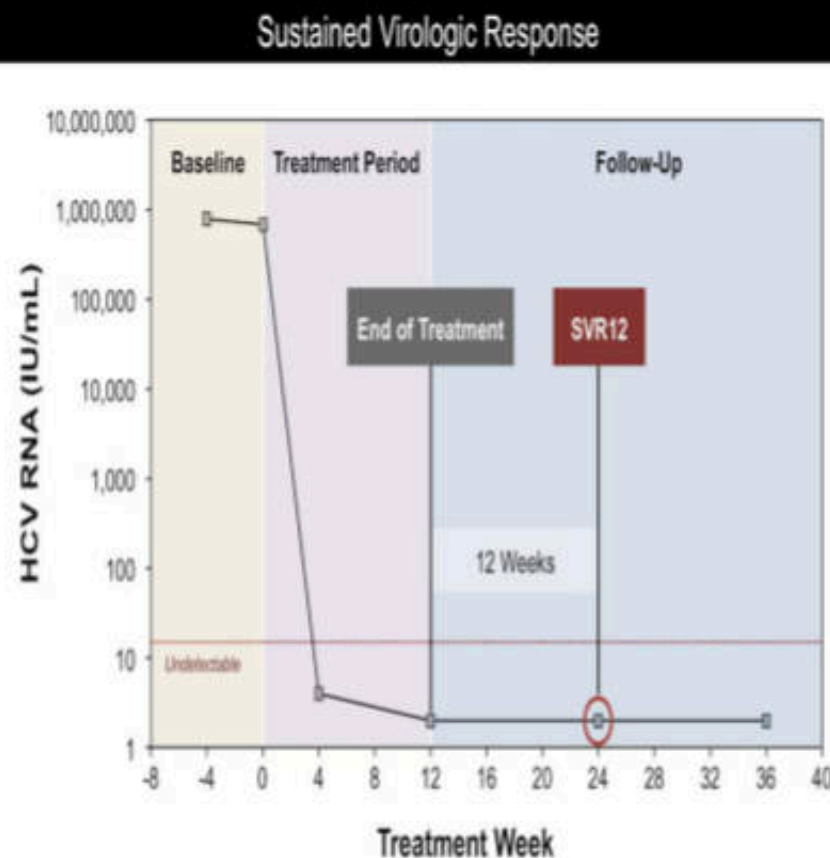
1. **prevent the complications:** hepatic necroinflammation, fibrosis, cirrhosis, decompensation of cirrhosis, HCC, severe extra-hepatic manifestations and death
2. **Improve quality of life**
3. **Prevent onward transmission of HCV**

- **Endpoint of therapy: Sustained viral response (SVR)**

- undetectable HCV RNA in serum or plasma by a sensitive assay (LLD ≤ 15 IU/ml) 12 weeks (SVR12) or 24 weeks (SVR24) after the end of treatment

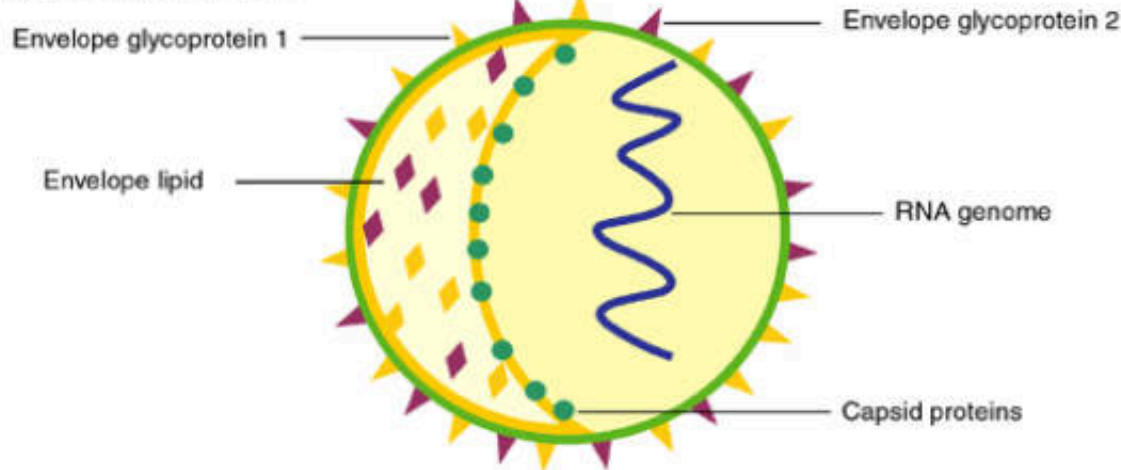
“Sustained viral response (SVR)”

- Surrogate marker for HCV cure
- SVR corresponds to a definitive cure of HCV infection in more than 99%
- Undetectable HCV RNA (HCV RNA <15 IU/mL) after the end of treatment for 12 weeks (**SVR12**) and/or 24 weeks (**SVR24**) **SVR24**

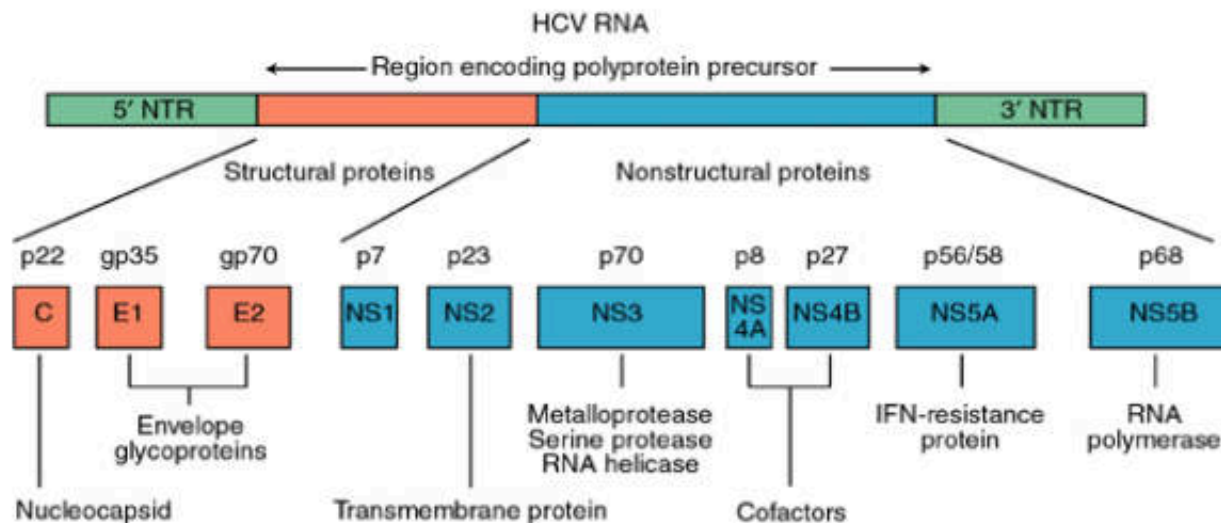


Hepatitis C virus (HCV) Structure

a Model structure of HCV



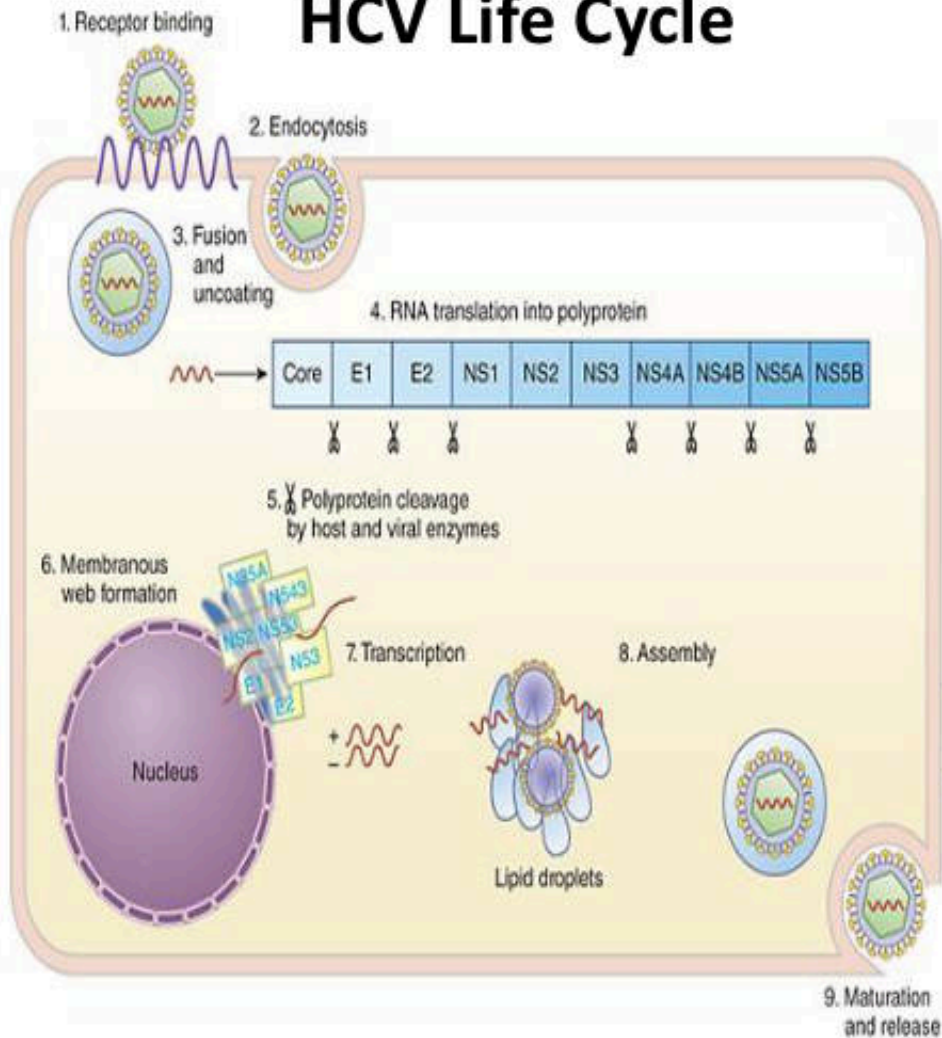
b Proteins encoded by the HCV genome



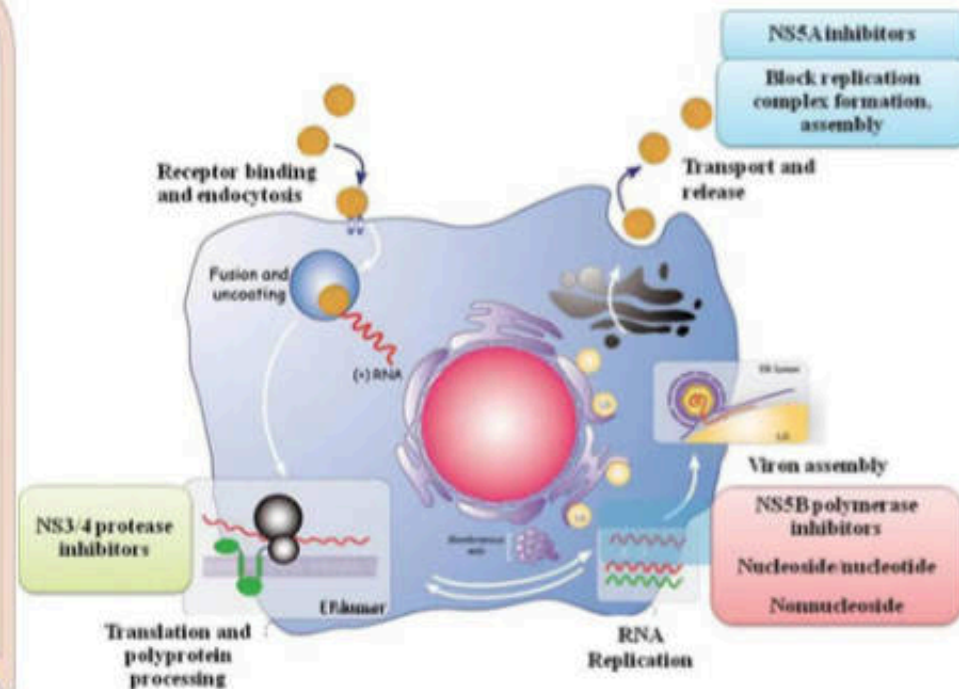
- genus Hepatitis C family *Flaviviridae*
- positive-sense, single-stranded **RNA** virus
- Contains 9024 to 9111 nucleotides (dependent on the genotype)
- **Structural proteins:** core, proteins E1 & E2
- **Non-structural proteins:** NS1(p7), NS2, NS3, NS4A, NS4B, NS5A, NS5B

The HCV Life Cycle and Drug targets

HCV Life Cycle

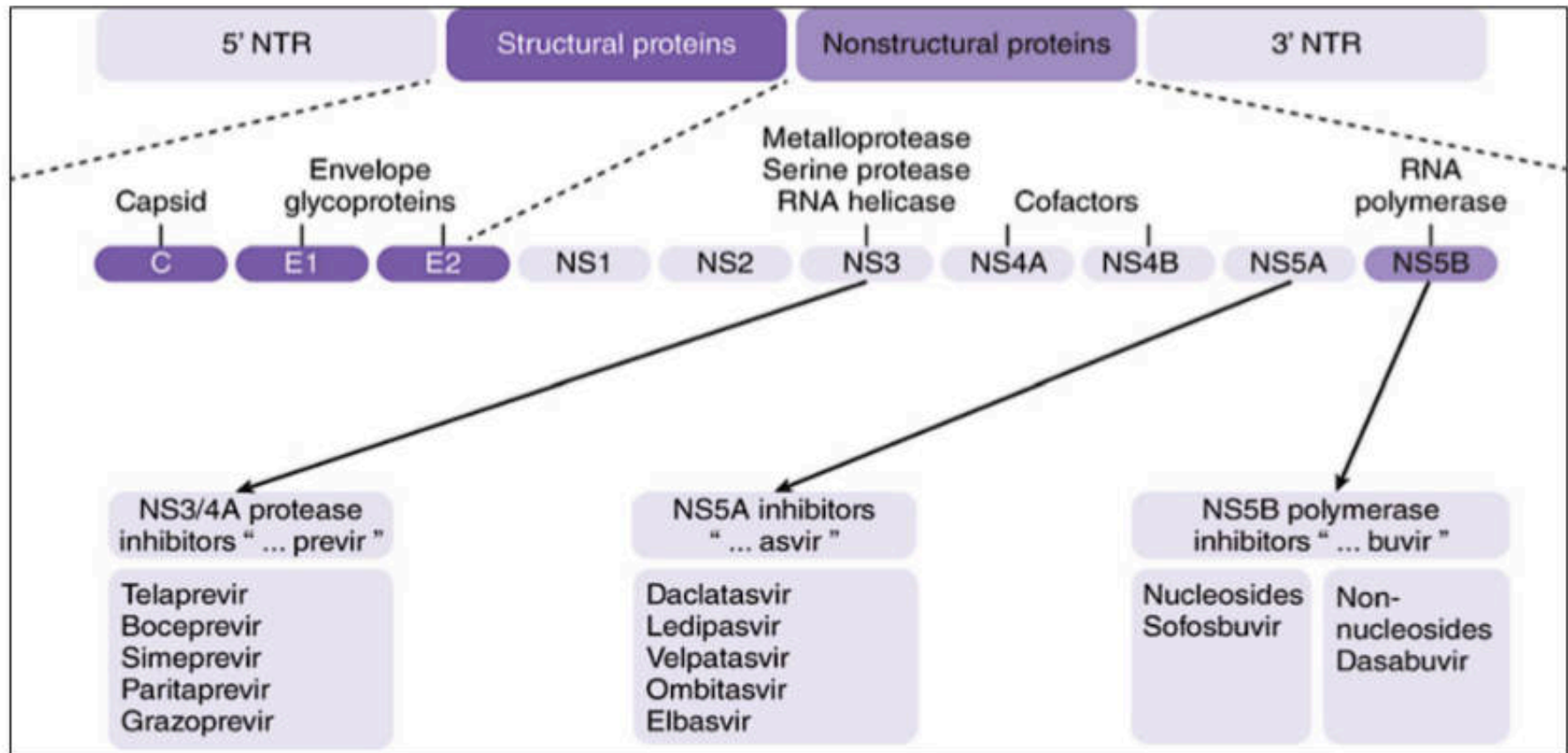


Targets for Direct-Acting Antiviral Drugs (DAAs)



HCV genome and potential targets of drug action

Knowing of structures of HCV proteins has permitted development of targeted inhibitors of HCV replication and revolutionized pharmacotherapy of HCV infection



C, E1, E2, protein products of specific genes; NS, nucleoside inhibitors; Non-NS, non-nucleoside inhibitors.

Antiviral Therapies for HCV Infection

1. Standard Treatments

- Interferon (IFN), PEGylated IFN $\pm$ Ribavirin (RBV)
- Non-specifically inhibit HCV replication
- 24-48 weeks of therapy
- severe side effects

2. Direct-Acting Antivirals (DAAs)

- NS3/4A Protease Inhibitors
- NS5B Nucleoside/nucleotide inhibitors
- NS5A Inhibitor
- Specifically inhibit HCV replication
- 12-24 weeks of therapy

Limitation of Interferon & Ribavirin



PEGIFN-ALFA: AN INHIBITOR OF HCV REPLICATION THAT INDUCES SPECIFIC HOST GENES WITH ANTIVIRAL FUNCTIONS



RIBAVIRIN: USED IN COMBINATION TO IMPROVE SVR RATES; BOOSTS EFFICACY SVR FROM 10% TO 50%

Ribavirin

Mechanism of actions: not been fully elucidated, possibly mechanisms

- (1) inhibiting the HCV RdRp
- (2) depleting GTP (and thus nucleic acid synthesis in general) through inhibition of inosine 5'-monophosphate dehydrogenase
- (3) enhancing viral mutagenesis
- (4) converting the T-helper cell phenotype from 2 to 1
- (5) inducing IFN-stimulated genes
- (6) modulating natural killer cell response

Inhibits the replication of wide range DNA and RNA viruses of influenza A and B, parainfluenza, respiratory syncytial virus, paramyxoviruses, HCV, and HIV-1

PKs: oral bioavailability 45–64%, high-fat meals increase absorption, large Vd, eliminate by renal

ADRs: hemolytic anemia (dose-dependent), depression, fatigue, irritability, rash, cough, insomnia, nausea, and pruritus

Teratogenic and embryo toxic in animals

Contraindications of interferon and ribavirin

ผู้ป่วยตับแข็งที่เป็นมากแล้ว (decompensated cirrhosis, CTP score > 6)

มีประวัติแพ้ยา **interferon** และ/หรือ **ribavirin**

ภาวะซึมเศร้ารุนแรงที่ยังควบคุมไม่ได้

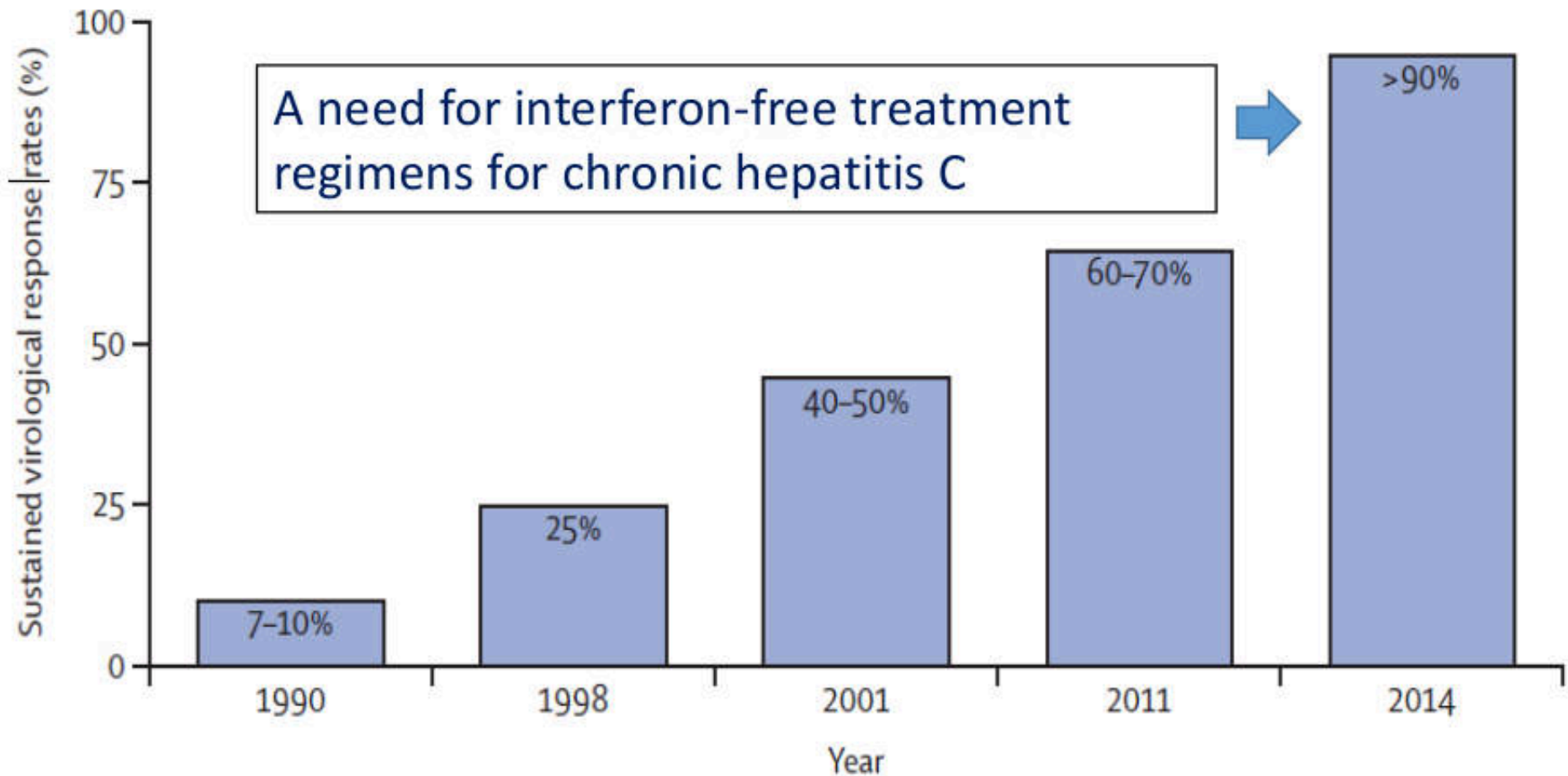
ตั้งครรภ์หรือไม่เต็มใจที่จะยินยอมในการคุมกำเนิด

ได้รับการผ่าตัดปลูกถ่ายอวัยวะ ยกเว้นการผ่าตัดเปลี่ยนตับ

มีโรคที่เป็นข้อห้ามต่อการใช้ยา **interferon**

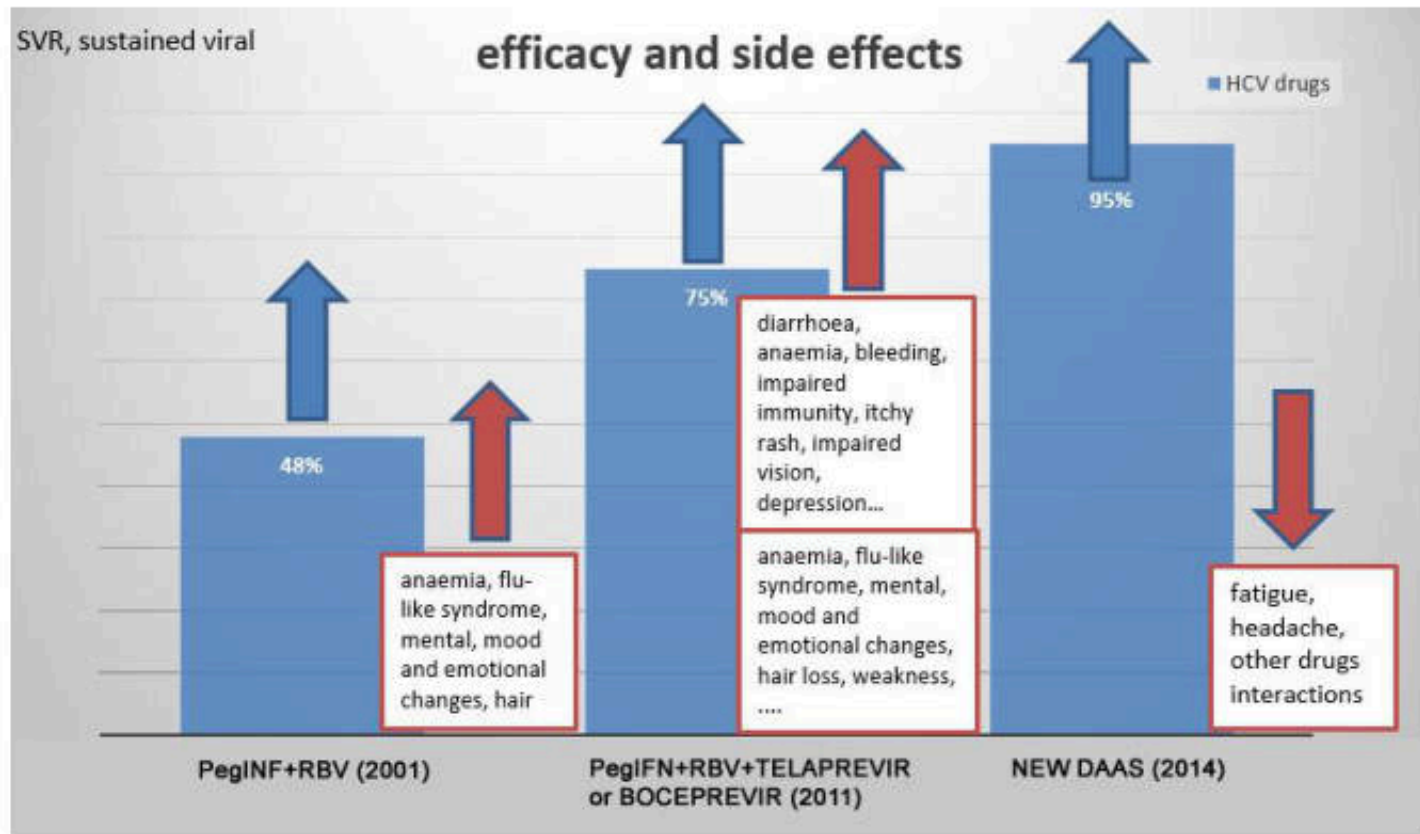
มีโรคประจำตัวที่ยังควบคุมรักษาได้ไม่ดี เช่น ความดันโลหิตสูง เบาหวาน โรคหัวใจ และหลอดเลือด ภูมิคุ้มกันบกพร่อง และโรคไตเรื้อรัง เป็นต้น

Evolution of HCV Treatments



Treatment of anti-hepatitis C with DAAs

- Oral specific for treatment of HCV and use for short time.
- Less ADR as compared with IFN
- Numerous and complex drug-drug interactions are possible with HCV DAAs
- Patients should be education on the importance of adherence therapy



Clinical Implications of HCV Resistance



High genetic barrier
to resistance:

Nucleotide NS5B polymerase
inhibitors
three or more resistance-
associated variants
(**RAVs**) are required for full
resistance

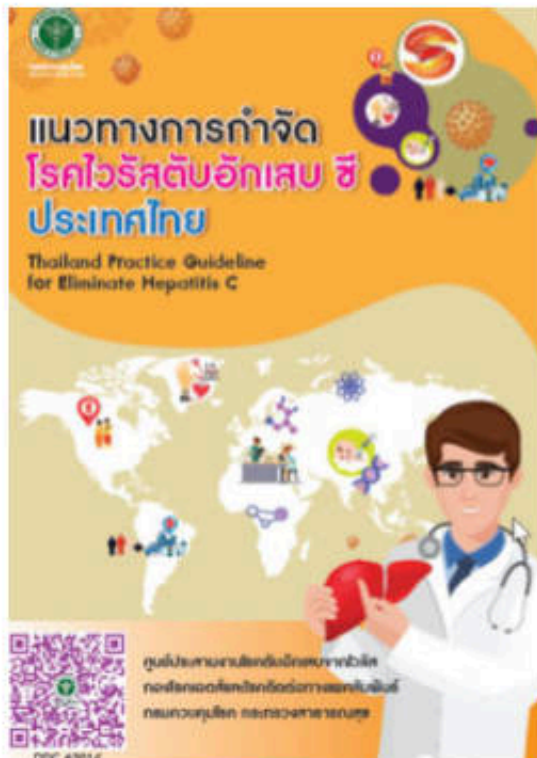


Low genetic barrier
to the development
of resistance:

Non-nucleoside NS5B
polymerase inhibitors, NS3
protease inhibitors, and
NS5A inhibitors
only one or two amino acid
substitutions result in lack of
drug efficacy

HCV
treatment
requires a
combination
of agents

Update guideline of HCV treatment



Guidelines



JOURNAL
OF HEPATOLOGY

EASL Recommendations on Treatment of Hepatitis C 2018*

European Association for the Study of the Liver*

Summary

Hepatitis C virus (HCV) infection is a major cause of chronic liver disease, with approximately 71 million chronically infected individuals worldwide. Clinical care for patients with HCV-related liver disease has advanced considerably thanks to an enhanced understanding of the pathophysiology of the disease, and because of developments in diagnostic procedures and improvements in therapy and prevention. These European Association for the Study of the Liver Recommendations on Treatment of Hepatitis C describe the optimal management of patients with acute and chronic HCV infection in 2018 and onwards.

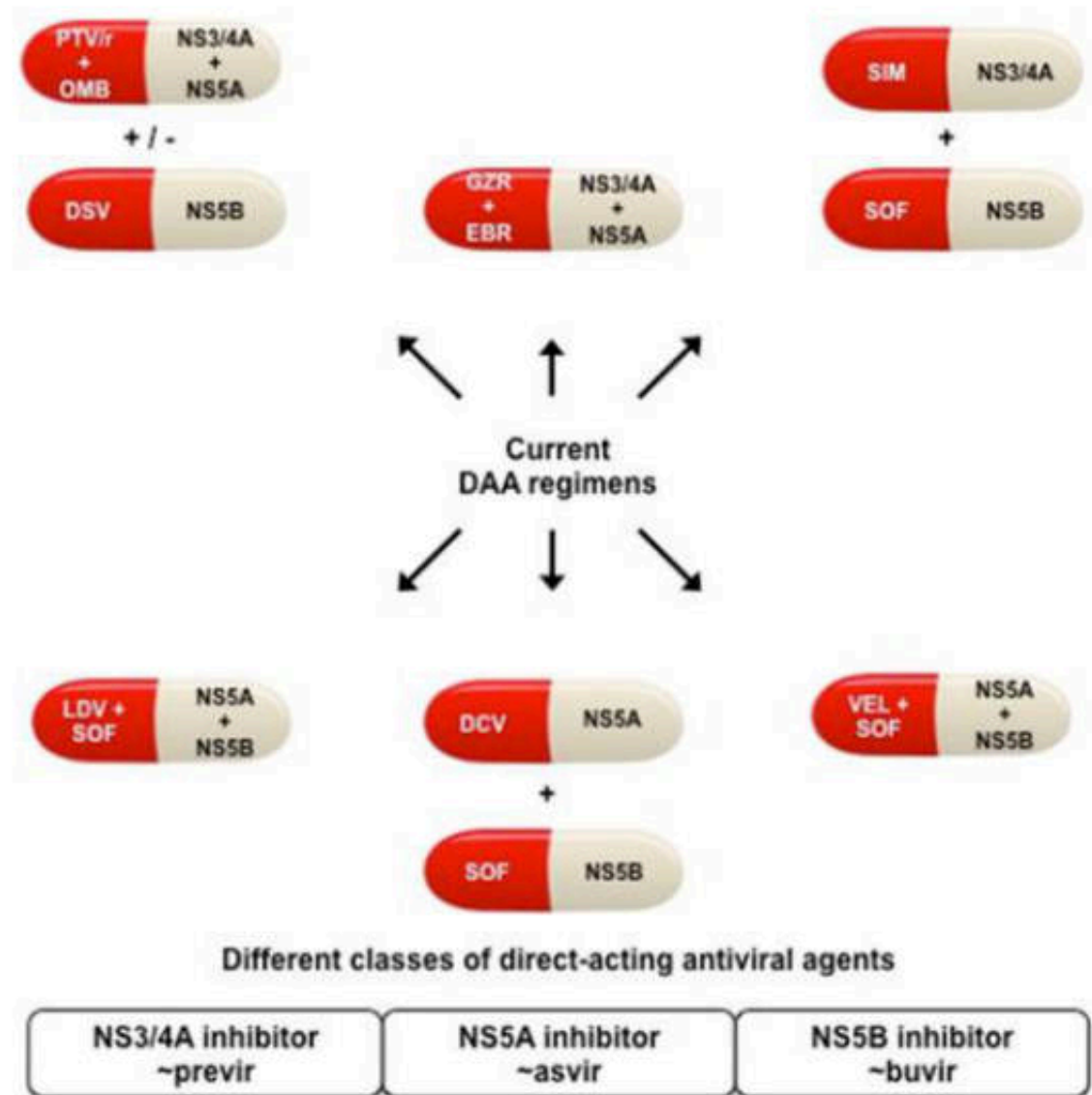
© 2018 European Association for the Study of the Liver. Published by Elsevier B.V. All rights reserved.

Serious and portal hypertension is reduced after an SVR. Recent data suggest that the risk of HCC and liver-related mortality is significantly reduced, but not eliminated, in patients with cirrhosis who clear HCV compared to untreated patients and non-sustained virological responders, especially in the presence of reduction of liver nodularity, such as the metabolic syndrome, harmful alcohol consumption and concurrent hepatitis B virus (HBV) infection.^{1,2} HCV is also associated with a number of extra-hepatic manifestations and viral elimination induces reversal of most of them with reduction of all-cause mortality.^{3,4}

These EASL Recommendations on Treatment of Hepatitis C are intended to assist physicians and other healthcare providers, as well as patients and other interested individuals, in the clinical decision-making process, by describing the current optimal management of patients with acute and chronic HCV infection.

Clinical Microbiology
and Infection.
2016;22(10): 846-852

Oral direct-
acting antiviral
regimens for
hepatitis C
virus treatment
with all
different
classes.



Abbreviations: HCV, hepatitis C virus; DAA, direct-acting Antivirals; OMB, ombitasvir; DSV, dasabuvir; SIM, simeprevir; SOF, sofosbuvir; LDV, ledipasvir; DCV, daclatasvir; GZR, grazoprevir; EBR, elbasvir; VEL, velpatasvir; PTV/r, ritonavir boosted paritaprevir.

Approved HCV drugs in the Thailand

- สูตรการรักษาและขนาดยาระยะเวลาการรักษา 12 wks. (+/- HIV)
- No cirrhosis:
 - Sofosbuvir 400mg+ Velpatasvir 100 mg (OD)
- With cirrhosis
 - Sofosbuvir 400 mg ++ Velpatasvir 100 mg (OD) + Ribavirin bid

Dose Ribavirin (BW)

Dose	Bodyweight	Frequency
1000 mg/d	< 75 kg	2 times/day
1,200 mg/d	>75 kg	2 times/day

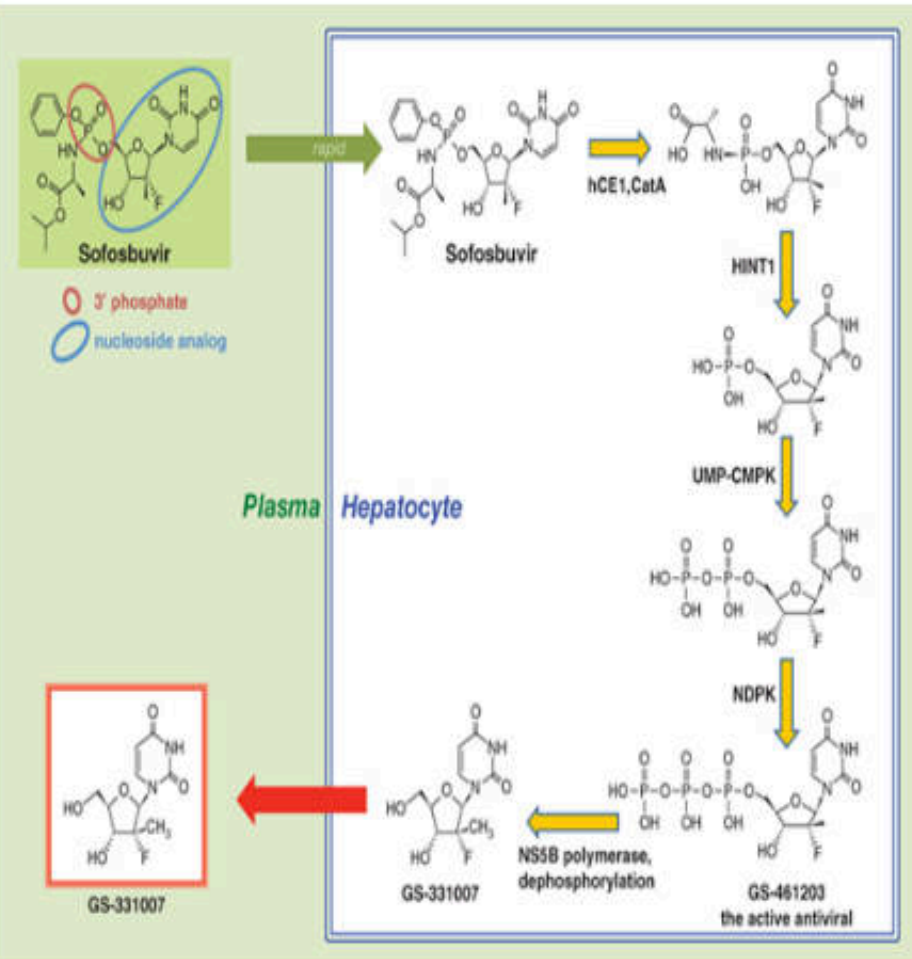
Sofosbuvir

Prodrug based on a uridine analogue

Mechanism of action: Metabolized to an active form (GS-331007) that is generated in the host mammalian cell, inhibits HCV RNA polymerase

Provides a high barrier to the development of resistance

- Recently, the S282T RAV has been reported in patients who relapsed to sofosbuvir-based treatment



Sofosbuvir (cont.)

- Dose 400 mg taken once daily, high-fat meal increases AUC by 67%–91
- $T_{1/2}$ of sofosbuvir: 0.4 h, GS-331007: 27 h.
- Pharmacokinetics of both sofosbuvir and GS-331007 affected by **renal impairment**
 - should not be used in patients with $eGFR < 30 \text{ mL/min/1.73 m}^2$
- Not a CYP substrate, inhibitor, or inducer → with low potential for drug interaction
- a substrate for the efflux transporters P-gp and BCRP → should not be used with potent inducers
- **THERAPEUTIC USES:** active against all HCV genotypes, used in combination with other DAAs
- **ADRs:** fatigue, headache, insomnia, pruritus, bradycardia (taking with amiodarone)

Example of drug interaction with sofosbuvir

Concomitant Drug Class: Drug Name	Effect on Concentration ^b	Clinical Comment
Anticonvulsants: carbamazepine phenytoin phenobarbital oxcarbazepine	↓ sofosbuvir ↓ GS-331007	Coadministration of SOVALDI with carbamazepine, phenytoin, phenobarbital or oxcarbazepine is expected to decrease the concentration of sofosbuvir, leading to reduced therapeutic effect of SOVALDI. Coadministration is not recommended.
Antimycobacterials: rifabutin rifampin rifapentine	↓ sofosbuvir ↓ GS-331007	Coadministration of SOVALDI with rifabutin or rifapentine is expected to decrease the concentration of sofosbuvir, leading to reduced therapeutic effect of SOVALDI. Coadministration is not recommended. SOVALDI should not be used with rifampin, a potent intestinal P-gp inducer [See <i>Warnings and Precautions</i> (5.2)].
Herbal Supplements: St. John's wort (<i>Hypericum perforatum</i>)	↓ sofosbuvir ↓ GS-331007	SOVALDI should not be used with St. John's wort, a potent intestinal P-gp inducer [See <i>Warnings and Precautions</i> (5.2)].
HIV Protease Inhibitors: tipranavir/ritonavir	↓ sofosbuvir ↓ GS-331007	Coadministration of SOVALDI with tipranavir/ritonavir is expected to decrease the concentration of sofosbuvir, leading to reduced therapeutic effect of SOVALDI. Coadministration is not recommended.

Drug interaction with SOF/VEL

แนวทางการตรวจคัดกรองและรักษาโรคไวรัสตับอักเสบ บี และ ซี สำหรับแพทย์เวชปฏิบัติทั่วไป พ.ศ.
2567

ยา	Contraindication	Potential interaction (Require dose adjustment and monitoring)
ยาด้านไวรัสเอชไอวี	Efavirenz, Nevirapine, Lopinavir	Rilpivirine, Atazanavir, Ritonavir, Darunavir
ยาลดกรด	PPIs (ยกเว้น Omeprazole 20 mg)	Omeprazole 20 mg* H2RA**, Antacids***
ยากันชัก	Carbamazepine, Oxcarbazepine, Phenobarbital, Phenytoin	
ยาหลอดเลือดหัวใจและยา ลดไขมัน	Amiodarone	Statins Digoxin, Dabigatran, Ticagrelor, Carvedilol, Diltiazem

* หากจำเป็น ควรให้ SOF/VEL พร้อมอาหาร ที่เวลา 4 ชั่วโมงก่อน PPI และไม่ใช่ Omeprazole ขนาดสูงกว่า 20 mg/d

** หากจำเป็น ควรให้ SOF/VEL พร้อมกับ H2RA หรือห่างกับ H2RA อย่างน้อย 12 ชั่วโมง และไม่ใช่ H2RA ขนาดสูงกว่าขนาดที่เทียบเท่ากับ Famotidine 40 mg BID

***ควรให้ SOF/VEL ห่างจากยา Antacids อย่างน้อย 4 ชั่วโมง

NS5A inhibitor

fixed dose combination product with sofosbuvir (SOF/VEL)

- **SOF/VEL: fixed-dose combination tablet (400 mg SOF +100 mg VEL) taken once daily**

PKs: absorption is pH dependent, 99.5% protein bound, T1/2: 15 h., should not used in eGFR <30 mL/min/1.73 m²

substrate for and weak inhibitor of Pgp and a weak inhibitor of BCRP and OATP1B1/1B3 substrate for CYP3A4, 2C8, and 2B6

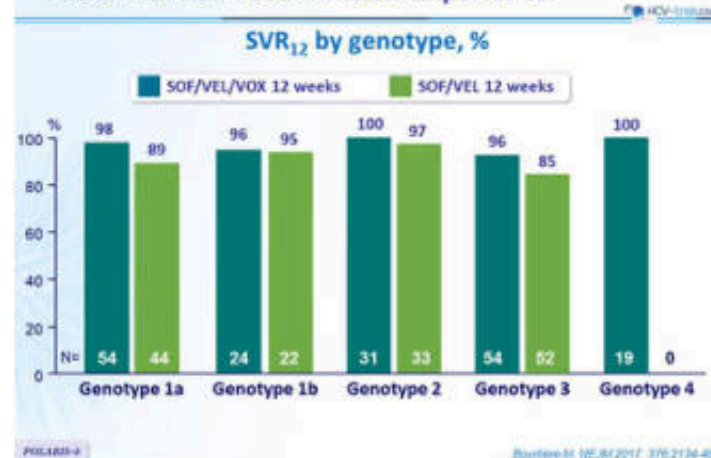
THERAPEUTIC USES: all HCV genotypes (1–6)

UNTOWARD EFFECTS: headache (22%), fatigue (15%), nausea (9%), asthenia (5%), and insomnia (5%)

Velpatasvir

sofosbuvir/velpatasvir/voxilaprevir
(SOF/VEL/VOX) fixed dose
combination

POLARIS-4 study: SOF/VEL/VOX vs SOF/VEL in genotypes 1 to 6 with non-NS5A inhibitor experience



Daclatasvir

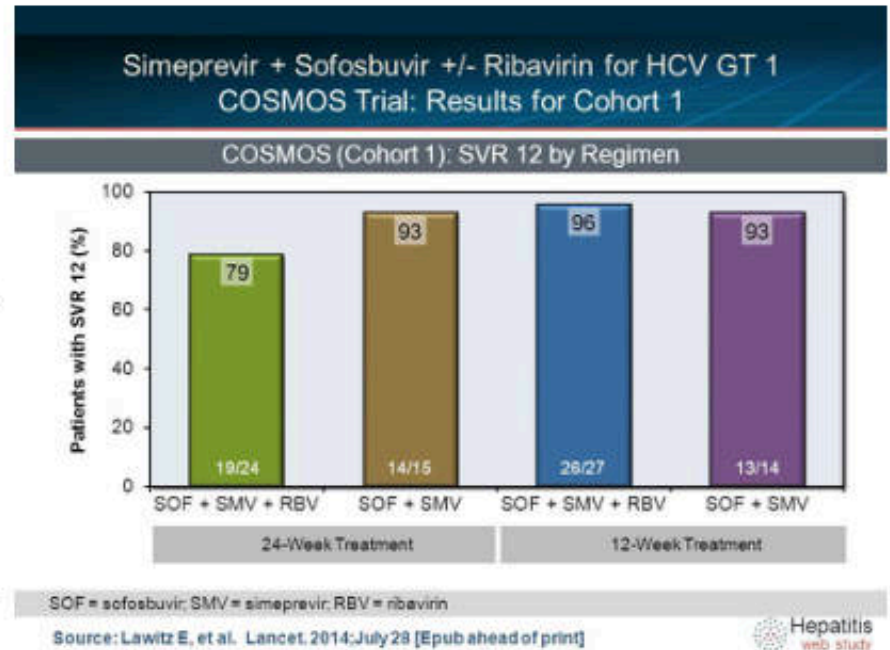
- NS5A inhibitor
 - binds to the N-terminus of NS5A and inhibits both viral RNA replication and virion assembly
- Resistance: the Y93H RAV
- PKs: Oral bioavailability 67%, high-fat, high-calorie meal reduces absorption (23%), highly protein bound (99%), T1/2: 12-15 h
- metabolized by CYP3A, substrate of P-gp → cannot be used with potent CYP3A inducers
- **THERAPEUTIC USES:**
 - US FDA-approved for use in combination with sofosbuvir in genotype 3
 - HCV with HIV coinfection regardless of HCV genotype
 - HCV with advanced liver disease
- Daclatasvir + Asunaprevir: Approved only in Japan
- **ADRS:** Well tolerated, headache and fatigue (most common)

Daclatasvir (DCV) dosing with Inducers and Inhibitors of CYP3A4

STRONG CYP3A INHIBITORS	MODERATE CYP3A INHIBITORS	STRONG CYP3A INDUCERS	MODERATE CYP3A INDUCERS
Decrease DCV Dose to 30 mg	Standard DCV Dose, 60 mg	DCV Contraindicated	Increase DCV Dose to 90 mg
Ritonavir-boosted atazanavir	Ritonavir-boosted darunavir	Rifamycins	Bosentan
Clarithromycin	Ritonavir-boosted lopinavir	St. John's Wort	Dexamethasone
Itraconazole/Voriconazole	Ciprofloxacin	Antiepileptics	Efavirenz
Ketoconazole	Diltiazem		Etravirine
Nefazodone	Erythromycin		Modafinil
Nelfinavir	Fluconazole		Nafcillin
Posaconazole	Fosamprenavir		Rifapentine
Telithromycin	Verapamil		

Simeprevir

- HCV NS3 protease inhibitor → prevent the formation of viral RNA
- Use in combination with sofosbuvir (SOF)
- **RESISTANCE:** R155K/D168E -emergent RAVs
- PKs: dosed as 150-mg tablet (taken once daily with food), bioavailability 62%, 99.9% protein bound, $T_{1/2}$: 10-13 h.
- Metabolized by CYP3A4, substrate for P-gp, MRP2, BCRP, OATP1B1/3, and OATP2B1, mild inhibitor of CYP1A2 and intestinal CYP3A
- Should not be used in patient with $eGFR < 30$ mL/min/1.73 m²
- **THERAPEUTIC USES:** genotype 1 or 4,
 - SVR is achieved in 97% of non-cirrhotic, treatment-naïve; the SVR rate is lower (83%) in cirrhotic patients
- **ADRs:** headache, fatigue, and nausea
- photosensitivity, rash, and itching (a sulfonamide moiety in structure)

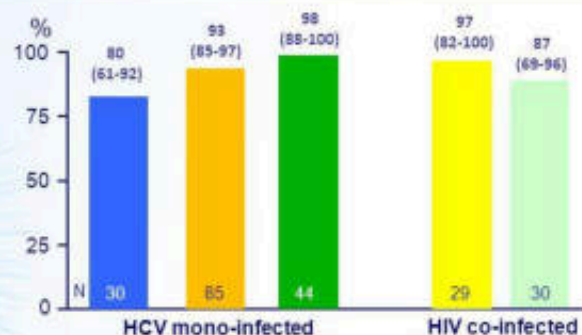


Grazoprevir/Elbasvir

ASAP-WORTHY Study: grazoprevir + elbasvir with or without ribavirin for genotype 1

SVR₁₂ (HCV RNA < 25 IU/ml), % (95% CI)

■ GZR + EBR + RBV 8W ■ GZR + EBR 12W ■ GZR + EBR 12W
■ GZR + EBR + RBV 12W ■ GZR + EBR + RBV 12W



discontinuation	1	3	0	0	2
glyc breakthrough	0	1	0	0	2
se	5	2	1	1	0

10/12 failure

Sulkowski M. Lancet 2

- Grazoprevir: NS3/4A protease inhibitor
 - Higher activity against more genotypes and higher barrier to the development of resistance than other
- Elbasvir: NS5A inhibitor
- Single fixed-dose combination tablet (**GRA100/ELB50**)
- US FDA approved for HCV genotypes 1a/1b and 4, in individuals with HIV coinfection
- ADRs: headache (17%), fatigue (16%), and nausea (9%–15%)

Caution of DAAs regimen

- **Drug-drug interactions** are possible with HCV DAAs and HIV-HCV coinfecting patients
- Patients should be educated on the importance of **adherence to therapy**

สูตรยา DAAs	ADRs during treatment
Sofosbuvir + velpatasvir	diarrhea 15-18% in patients receiving the triple combination and 5-7% of sofosbuvir and velpatasvir only
Glecaprevir + pibrentasvir	fatigue and headache
Sofosbuvir + ledipasvir	Fatigue and headache
Grazoprevir + elbasvir	fatigue, headache, and nausea, ALT elevations up to >5 times ULN
Ritonavir-boosted paritaprevir, ombitasvir + dasabuvir	pruritus, fatigue, nausea, asthenia and insomnia, Transient increases in indirect serum bilirubin

Drug-drug interactions between HCV DAAs and cardiovascular drugs

		SOF	SOF/VEL	SOF/VEL/VOX	GLE/PIB	GZR/EBR
Antiarrhythmics	Amiodarone	●	●	●	■	■
	Digoxin	◆	■	■	■	◆
	Vernakalant	◆	◆	◆	◆	◆
	Flecainide	◆	◆	◆	◆	◆
Beta-blockers	Atenolol	◆	◆	◆	◆	◆
	Bisoprolol	◆	◆	◆	◆	◆
	Carvedilol	◆	■	■	■	◆
	Propranolol	◆	◆	◆	◆	◆
Calcium channel blockers	Amlodipine	◆	◆	◆	◆	◆
	Diltiazem	◆	■	■	■	◆
	Nifedipine	◆	◆	◆	◆	◆
Hypertension and heart failure agents	Losartan	◆	◆	◆	◆	◆
	Doxazosin	◆	◆	◆	◆	◆
	Enalapril	◆	◆	■	■	◆

Table 4F. Drug-drug interactions between HCV DAAs and immunosuppressants.

	SOF	SOF/VEL	SOF/VEL/VOX	GLE/PIB	GZR/EBR
Azathioprine	◆	◆	◆	◆	◆
Cyclosporine	◆	◆	●	■	●
Etanercept	◆	◆	◆	◆	◆
Mycophenolate	◆	◆	◆	◆	◆
Sirolimus	◆	◆	■	■	■
Tacrolimus	◆	◆	■	■	■

Table 4G. Drug-drug interactions between HCV DAAs and antiplatelets and anticoagulants.

	SOF	SOF/VEL	SOF/VEL/VOX	GLE/PIB	GZR/EBR
Clopidogrel	◆	◆	◆	◆	◆
Dabigatran	◆	■	●	●	■
Ticagrelor	◆	■	■	■	■
Rivaroxaban	◆	■	■	■	■
Apixaban	◆	■	■	■	■
Edoxaban	◆	■	●	■	■
Warfarin	■	■	■	■	■

DAAs, direct-acting antivirals; EBR, elbasvir; GLE, glecaprevir; GZR, grazoprevir; PIB, pibrentasvir; SOF, sofosbuvir; VEL, velpatasvir; VOX, voxilaprevir.

Colour Legend

- ◆ No clinically significant interaction expected.
- Potential interaction which may require a dosage adjustment, altered timing of administration or additional monitoring.
- These drugs should not be co-administered.

Notes:

- o Some drugs may require dose modifications dependent on hepatic function. Please refer to the product label for individual drugs for dosing advice.
- o The symbol (green, amber, red) used to rank the clinical significance of the drug interaction is based on www.hep-druginteractions.org (University of Liverpool). For additional drug-drug interactions and for a more extensive range of drugs, detailed pharmacokinetic interaction data and dosage adjustments, refer to the above-mentioned website.

Pharmacokinetics of DAAs which predispose them to drug-drug interactions

Class drug	Absorption	Distribution	Metabolism	Elimination mainly	
NS3/4A Protease Inhibitors			substrate for	inhibitor or inducer	
First-Wave Boceprevir	↑ by food	PPB 75% P-gp inhibitor	CYP3A4/5 AKR	Moderate CYP3A inhibitor	Hepatic
Telaprevir	↑ by fatty meals	PPB 60-76% Moderate P-gp inhibitor	CYP3A4/5	Strong CYP3A inhibitor	Hepatic
Second-Wave Simeprevir	↑ by food	PPB 99% inhibitor of P-gp, OATP1B1 and MRP2	CYP3A	mildly inhibits intestinal CYP3A4 and CYP1A2	Hepatic
Asunaprevir	↑ by fatty meals	PPB 99% inhibitor of P-gp and OATP1B1	CYP3A	Moderate inhibitor of CYP2D6; weak inducer of CYP3A4	Hepatic
Paritaprevir	↑ by food	PPB 97-98-% substrate for P-gp and BCRP; inhibitor of OATP1B1	CYP3A4/5	ND	Hepatic
Grazoprevir	↑ by fatty meals	PPB 98.8% Transported by OATP1B1, P-gp. Inhibits BCRP	CYP3A	Weak inhibitor of CYP3A	Hepatic
NS5A Inhibitors					
Daclatasvir	↓ by fatty meals	PPB 99% inhibitor of P-gp and OATP1B1	CYP3A	ND	Hepatic
Ledipasvir	↓ by increasing gastric pH	PPB 99% substrate and an inhibitor of P-gp	Minimally metabolized	ND	as parent drug
Ombitasvir	↑ by food	PPB 99% substrate of P-gp and BCRP	Amide hydrolysis	ND	Hepatic
Elbasvir	↓ by fatty meals	PPB 99.9% Transported by, P-gp. Inhibits BCRP, P-gp	CYP3A	ND	Hepatic
Velpatasvir	↑ by fatty meals	PPB 99.5% Transported by and inhibits P-gp, BCRP, OATP1B1, OATP1B3	CYP2B6, CYP2C8, CYP3A4		Hepatic
NS5B Polymerase Inhibitors					
Nucleoside Inhibitors					
Sofosbuvir	Food ↔	PPB 61-65% Substrate of P-gp and BCRP	Prodrug is phosphorylated to active-form	ND	Renal
Non-Nucleoside Inhibitors					
Dasabuvir	↑ by food	PPB 99% substrate of P-gp and BCRP	CYP2C8 CYP3A, CYP2D6	ND	Hepatic

AKR: aldo keto reductase; BCRP: breast cancer resistance protein; CYP: cytochrome P450; DAA: direct-acting antiviral agent; MRP: multidrug resistant protein; ND: no data; OATP: organic anion transporting polypeptide; P-gp: P-glycoprotein; PPB: plasma protein binding. ↑ increase; ↓ decrease; ↔ nil

NATIONAL LIST OF ESSENTIAL

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5.3.3. Viral hepatitis

5.3.3. Viral hepatitis				
1	Tenofovir alafenamide	TAF	tab (เฉพาะ 25 mg)	✓
2	Tenofovir disoproxil fumarate	TDF	tab	✓
3	Lamivudine	3TC	syr	✓
4	Entecavir	ETV	tab (เฉพาะ 0.5 mg), oral sol	✓
5	Sofosbuvir + Velpatasvir		tab (เฉพาะ 400 mg + 100 mg)	✓(2)
6	Ribavirin		cap/tab (เฉพาะ 200 mg)	✓(2)

Summary of Antihepatitis drugs

- Interferon is largely replaced due to toxicity
- HBV treatment is long-term suppression but not cure
- HCV treatment: DAAs achieve cure in most patients
- Tenofovir & entecavir are HBV first-line drugs
- During treatment assess: Renal function/Liver function/Drug–drug interactions
- Drug adherent are the most important for therapy

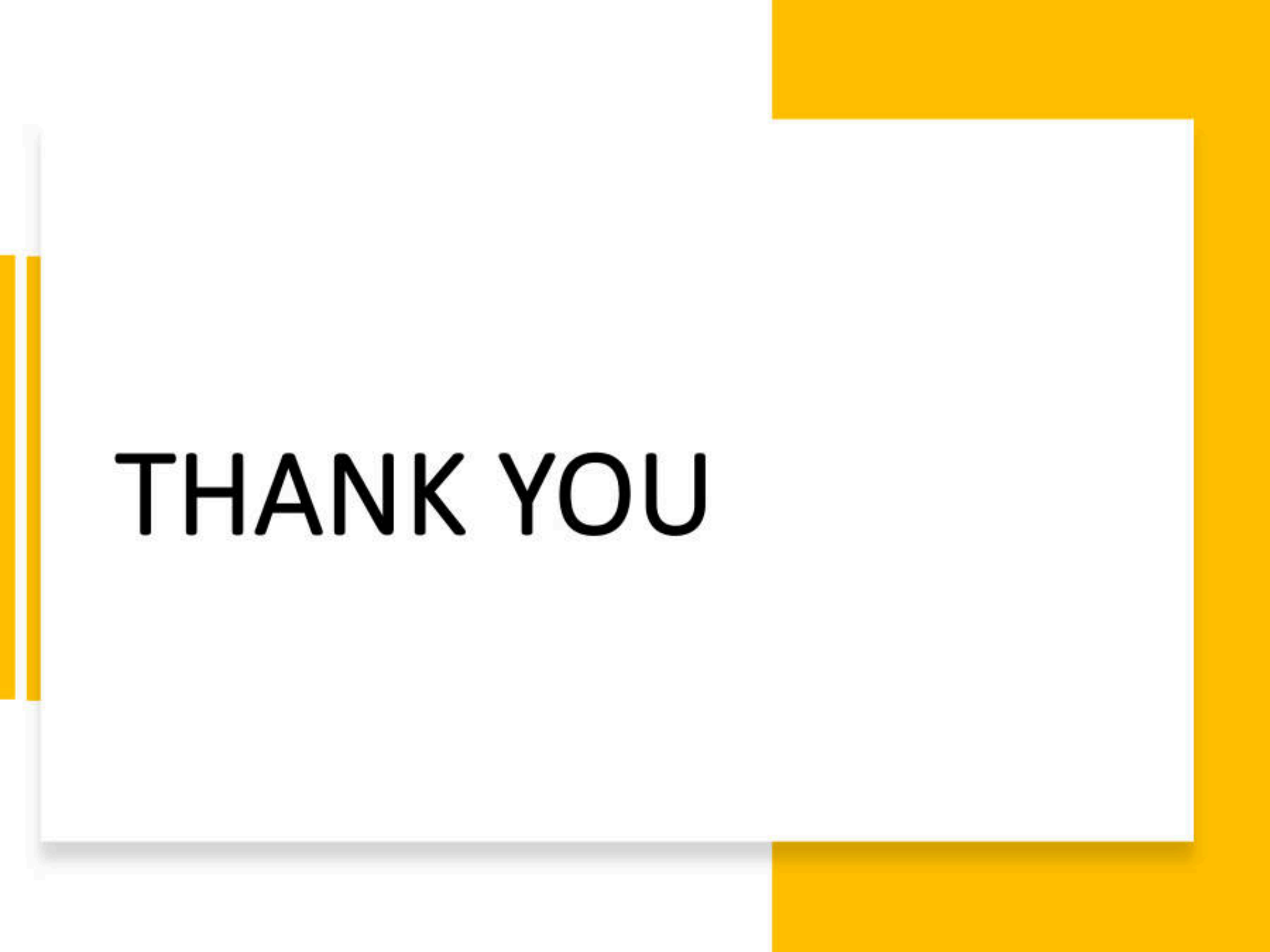
Quiz

1. What is the primary goal of treatment in patients with chronic hepatitis C infection?

- A. Suppression of HCV RNA during therapy
- B. Reduction of ALT levels only
- C. Achievement of sustained virologic response (SVR)
- D. Prevention of viral resistance

2. What is a major advantage of DAA regimens compared with interferon-based therapy ?

- A. Shorter half-life
- B. Higher rates of resistance
- C. Better tolerability and oral administration
- D. Requirement for combination with ribavirin

The image features a clean, modern aesthetic with a white background. In the top-left corner, there is a yellow square. In the top-right corner, there is a yellow rectangle. In the bottom-left corner, there is a yellow rectangle. In the bottom-right corner, there is a yellow square. The text "THANK YOU" is centered in the white area.

THANK YOU